GS FOUNDATION 2024-25

GEOGRAPHY - 20

Water (Oceans)

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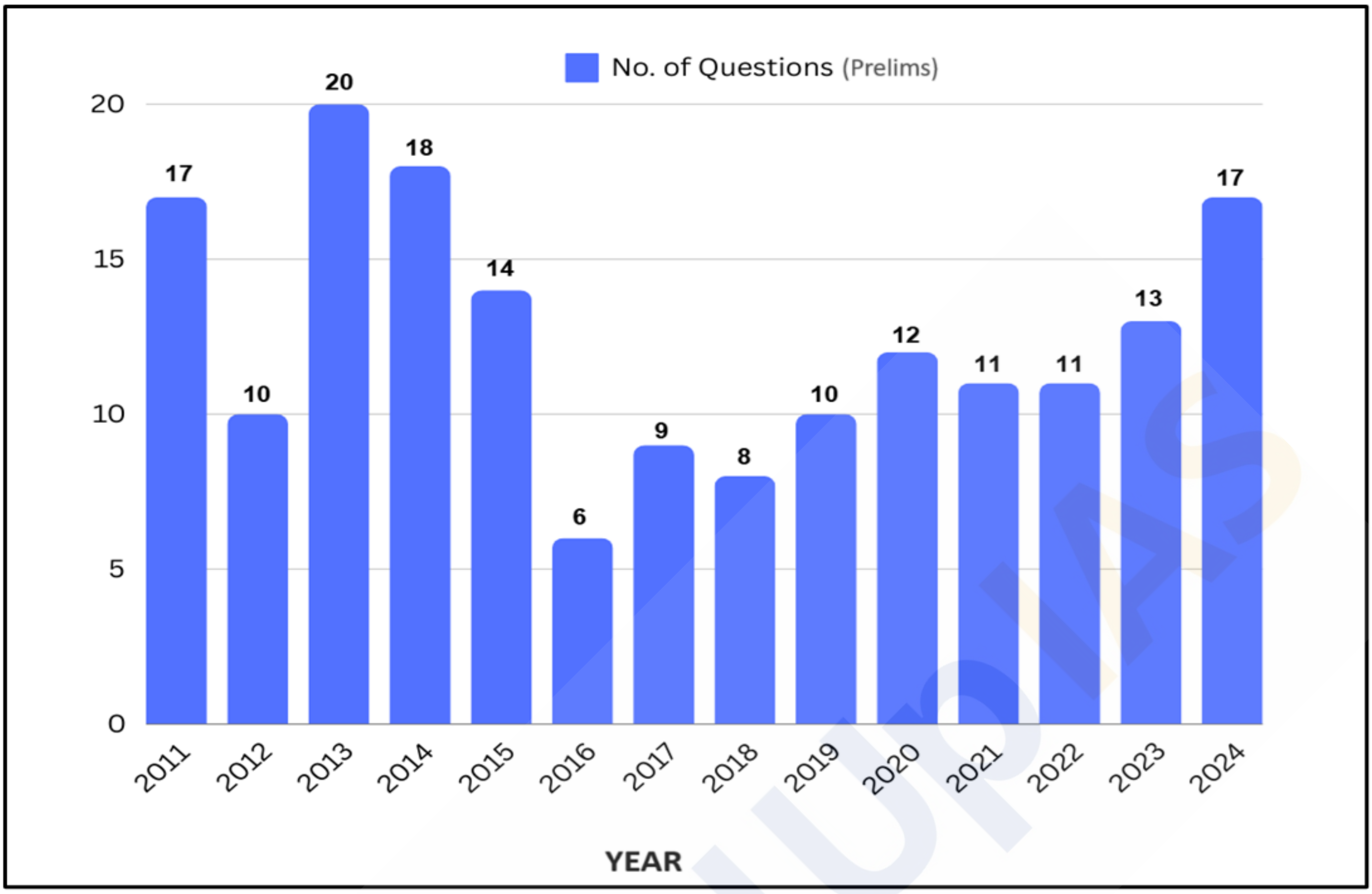

Topic-wise Trend Analysis (2011 – 2024)

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Water (Oceans)

Introduction:

Water is often described as the **essence of life,** a statement that holds profound truth. All life forms on Earth, from the simplest microorganisms to the most complex beings, are dependent on water. Our planet is fortunate to have an abundance of this vital resource, earning it the name **"Blue Planet."** In the vast expanse of our solar system, Earth is unique in its wealth of water, making it a rare and precious commodity. The absence of water on other celestial bodies, such as the Sun and elsewhere in our solar system, underscores the privilege of our existence on a water-rich planet.

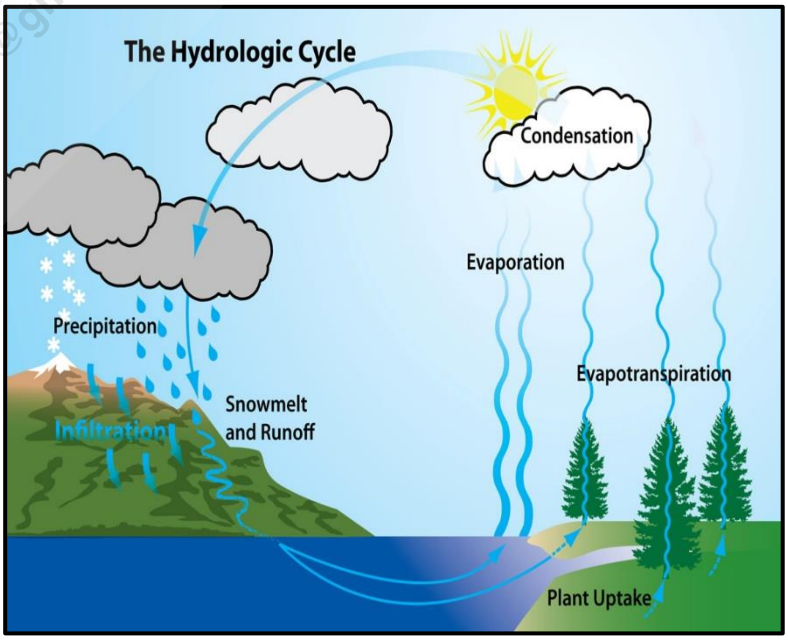
Covering approximately **71% of the Earth's surface with oceans,** water exists in three states - **Solid** (ice), **Liquid**, and **Gas** (water vapor). Oceans are indispensable to life on Earth. They **produce around 50% of the world's oxygen** and **absorb more than 50% of the carbon dioxide** from the atmosphere. Oceans play a key role in regulating the Earth's climate by distributing heat from the equator to the poles.

Hydrological Cycle:

Hydrology is the **scientific study of water's movement on the Earth's surface.** The hydrological cycle, also known as the **water cycle,** describes the **continuous movement of water in different states**—solid, liquid, and gas—through various processes such as evaporation, condensation, precipitation, and runoff. This process has been ongoing for billions of years and is fundamental to the existence of life on our planet. While the cycle **ensures that water is constantly renewed,** its distribution is uneven across the globe. Some regions are blessed with abundant water resources, while others face severe scarcity.

Key Processes in the Hydrological Cycle:

- **Evaporation -** Evaporation is the process by which **water transforms from a liquid state to a gaseous state** due to increased temperature and pressure. It is a fundamental part of the water cycle, with water primarily evaporating from oceans and other water bodies.



- **Condensation -** Condensation occurs when **water vapor in the atmosphere cools and changes back into liquid form,** leading to cloud formation. This process is the opposite of evaporation and requires low temperatures to occur.
- **Precipitation -** Precipitation happens when **accumulated water in the atmosphere falls back to the Earth's surface** in various forms, including rain, snow, sleet, fog, and hail. Gravity plays a key role in this process.
- **Evapotranspiration -** Evapotranspiration refers to the combined process of water being transferred from the land to the atmosphere through both **evaporation from the soil and transpiration from plants.** This process accounts for about **10%** of the atmospheric water vapor.

- **Transpiration -** Transpiration is the process by which **water is absorbed by plant roots and later evaporated from the plant's surface,** contributing to the atmospheric moisture.
- **Runoff -** Runoff is the portion of **precipitation that does not infiltrate the soil but instead flows** over the land surface, eventually reaching rivers, lakes, and oceans. Runoff is a major cause of soil erosion and can also carry pollutants, leading to water contamination.
- **Infiltration -** Infiltration is the process by which **water is absorbed into the soil.** This water can remain in the soil for extended periods or gradually evaporate. Infiltrated water is crucial for plant growth, especially in areas with abundant vegetation.
- **Groundwater Hydrology -** Groundwater refers to water that exists below the Earth's surface, filling the spaces within soils and geological formations. The **flow of groundwater is influenced by the soil's porosity and permeability,** and it plays a vital role in the hydrological cycle.
- **Aquifer -** An aquifer is an **underground layer of permeable rock or unconsolidated materials that can store and transmit groundwater.** Water from aquifers is often extracted using wells, and the study of these processes is known as hydrogeology.
- **Porosity and Permeability:**
    - **Porosity -** Refers to the **amount of open space within rocks or soil,** which can be between grains or within cracks and cavities.
    - **Permeability -** Measures **how easily water can flow through a porous material,** influenced by the size, shape, and sorting of sediment particles.

Relief of the Ocean Floor:

Ocean relief features are distinct from continental features due to the **younger age of the oceanic crust,** which is less than 60-70 million years old, compared to the Proterozoic age of continental features, which are over 1 billion years old.

Historically, there are four named oceans - the Atlantic, Pacific, Indian, and Arctic. Recently, the **Southern (Antarctic)** Ocean has been recognized as the **fifth ocean.** Among these, the Pacific, Atlantic, and Indian Oceans are considered the three major oceans.

Ocean relief is also shaped by **tectonic, volcanic, erosional,** and **depositional** processes and their interactions. The ocean floor is a complex and varied landscape. It features a range of geological formations including **mountains, basins, plateaus, ridges, canyons,** and **trenches.** These underwater landforms, collectively known as submarine relief, play a crucial role in shaping the ocean's geography.

Major Relief Features:

Continental Shelf: {source rich region}

- The continental shelf is a **shallow water area surrounding the main continental landmass,** geologically considered a submerged extension of the continent under oceanic waters.
- Typically, it is **about 100 fathoms deep and gently slopes** towards the ocean at an average rate of 20 meters per Kilometer, with the **angle of slope varying between 1° and 3°.**

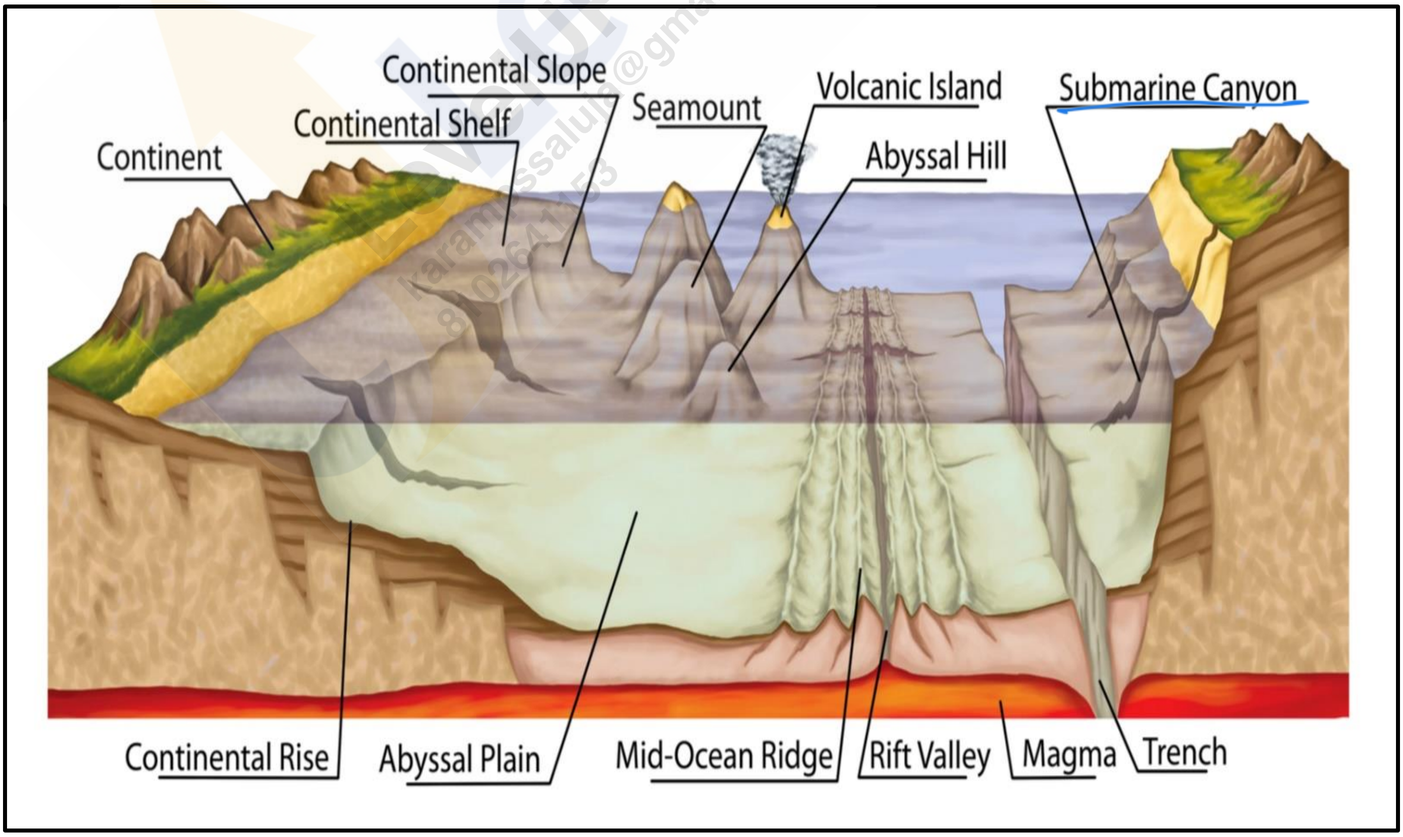
- The shelf typically ends at a steep slope known as the **shelf break.** The width of continental shelves varies across oceans, averaging about **80 km.** They are almost absent or very narrow along certain margins, such as the coasts of Chile and the west coast of Sumatra. Conversely, the **Siberian shelf in the Arctic Ocean, the world's largest, stretches up to 1,500 km in width.**
- Continental shelves are covered with **varying thicknesses of sediments deposited** by rivers, glaciers, wind, waves, and currents. These sedimentary deposits, accumulated over long periods, are significant sources of **fossil fuels.**

Significance of Continental Shelves:

- Continental shelves are **economically and ecologically significant.** They are among the most accessible parts of the ocean, providing conditions for the growth of millions of **plankton** and **microorganisms** through sunlight penetration. This makes them **excellent breeding grounds for fish and the richest fishing grounds** in the world. Marine food sources almost entirely come from continental shelves. Additionally, they are sources of fossil fuels and the formation of metallic and non-metallic ores.

Continental Slopes:

- The continental slope is the **inclined region** between the outer edge of the **continental shelf** and the **deep ocean floor.** This area is often **intersected by submarine canyons** in various locations. The continental slope marks the seaward boundary of the continental shelf and features a **gradient** that typically ranges from **2 to 5 degrees.** It spans depths from approximately **180 meters to 3,600 meters.**
- In certain areas, such as off the shore of the Philippines, the continental slope can extend to even greater depths. Due to its steepness and the increasing distance from land, the continental slope has **minimal sediment deposits.** Consequently, **marine life is far less abundant here compared to the continental shelf.**
- At the base of the continental slope, a belt of sedimentary deposits accumulates, forming what is known as the **continental rise.** The width of the continental rise varies significantly, ranging from very narrow in some regions to as much as 600 kilometers in others.



Continental Rise:

- The continental rise is a **major depositional system in oceans,** consisting of thick sequences of continental material that accumulate between the continental slope and the abyssal plain. These **thick sediments result from three sedimentary processes - mass wasting, deposition** from contour currents, and the **vertical settling** of clastic and biogenic particles.
- The **steepness** of the continental rise is **lower than that of the continental slope,** gradually merging into the deep sea plain.

Deep-Sea Plains or Abyssal Plains:

- Abyssal plains are **extensive, smooth, and flat areas** that lie **between mid-oceanic ridges and continental margins.** These plains are where continental sediments that move beyond the margins are deposited. They are covered with **sediments such as silt and mud.** Abyssal plains are commonly found in the Atlantic Ocean and are unusual in the Pacific Ocean.
- It covers nearly **40% of the ocean floor.** The depths vary between **3,000 and 6,000 m.** These plains are covered with **fine-grained sediments** like **clay and silt.** It has extensive submarine plateaus, ridges, trenches, beams, and oceanic islands that rise above sea level in the midst of oceans.

Minor Relief Features:

Mid-Ocean Ridges:

- A mid-oceanic ridge consists of **two chains of mountains separated by a large depression** at a **divergent boundary.** Some peaks within these mountain ranges can rise as high as 2,500 meters, with a few even extending above the ocean's surface.
- It forms the **largest mountain systems on Earth.** The ridge systems can appear as **broad, plateau-like features with gentle slopes or as steep-sided, narrow mountains.** Being of tectonic origin, these oceanic ridges provide strong evidence supporting the theory of Plate Tectonics. **Iceland,** which lies along the mid-Atlantic Ridge, serves as a notable example.

Seamount:

- A seamount is a **mountain with pointed summits** rising from the seafloor but **not reaching the ocean surface.** Seamounts are **volcanic in origin** and can be 3,000-4,500 m tall.
- **The Emperor seamount,** an extension of the **Hawaiian Islands** in the Pacific Ocean, is a notable example.

Submarine Canyons:

- Submarine canyons are **long, narrow, and deep valleys or trenches** located on continental shelves and slopes with **steep vertical walls.** These physical features, which start from the continental shelf break and extend to the continental slopes, resemble continental canyons but are situated underwater.
- Unlike deep-sea trenches found mostly at plate boundaries, undersea **canyons are present along the slopes** of most continental margins.

Ganga canyon. Mahanadi canyon

Ocean Deeps or Trenches:

Deepest part of Trench ocean deep

- Ocean deeps, also known as trenches, are **depressions on the ocean floors** and represent the **deepest zones of ocean basins.** They are **generally located parallel to coasts with nearby mountains** or along islands, typically in **subduction zones** where oceanic plates converge.

Marine Trench has challenger deep (11 km)

Guyots:

- Guyots, also known as **tablemounts,** are **isolated underwater volcanic mountains** characterized by their **flat tops,** which are located more than 200 meters below the sea surface.
- These formations exhibit evidence of **gradual subsidence, eventually becoming flat-topped submerged mountains.** It is estimated that over 10,000 seamounts and guyots can be found in the Pacific Ocean alone.

Atoll:

- Atolls are **low islands in tropical oceans,** consisting of **coral reefs surrounding a central depression,** which may be part of the sea (lagoon) or enclose a body of fresh, brackish, or highly saline water.

Banks:

- Banks are **flat-topped elevations found in continental margins, resulting from erosional and depositional activities.** They are characterized by shallow depths that are still adequate for navigation.
- Notable examples include the **Dogger Bank in the North Sea** and the **Grand Bank off the coast of Newfoundland** in the north-western Atlantic. Banks are known for being some of the world's most productive fishing grounds.

Shoals:

- A shoal is a **detached elevation with shallow waters.** These features, projecting above the water with moderate heights, can pose significant hazards to navigation due to their shallow depths.

Reefs:

- Reefs are predominantly **organic structures formed by living or dead organisms** that create mounds or rocky elevations resembling ridges. **Coral reefs,** which are particularly common in the **Pacific Ocean, are associated with seamounts and guyots.**
- The largest reef in the world is located off the **Queensland coast of Australia.** Reefs can extend above the water's surface and are therefore often dangerous for navigation.

→ based on latitude
→ based on depth

Temperature of Ocean Waters:

Avg. temp → 4-5°C
Avg. ocean depth → 4 km

The temperature of ocean waters exhibits both spatial and vertical variations across different oceans. Unlike land, ocean waters heat up and cool down more slowly due to their high heat capacity. **The ocean takes in about 80 percent of the Earth's heat.** The **upper 10% of the ocean's depth holds more heat than the entire atmosphere combined.**

Factors Affecting Temperature Distribution:

- **Latitude -** Ocean surface temperatures decrease as one moves from the ~~equator~~ Tropical towards the poles. This is due to the Sun's rays striking the equator more directly, resulting in warmer waters there compared to higher latitudes.
- **Prevailing Winds -** The direction of prevailing winds, including Trade Winds and Westerlies, affects the temperature of ocean waters. These winds influence the distribution of heat across the ocean's surface.

At higher latitude → Enclosed sea will have lower temp. th

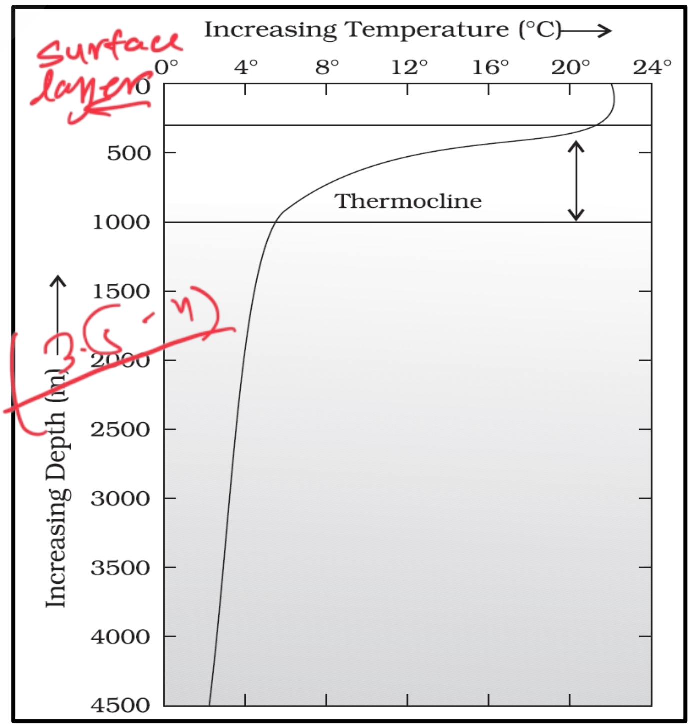
- **Unequal Distribution of Land and Water -** The Northern Hemisphere contains more landmass compared to the Southern Hemisphere. Consequently, the oceans in the Northern Hemisphere tend to be warmer than those in the Southern Hemisphere. Eg. Mediteran sea (Enclosed sea) → At lower latitude As more temp. than the avg. at this latitude.
- **Evaporation Rate -** Evaporation rates vary across different ocean regions. Warmer areas experience higher evaporation rates compared to cooler regions. This variation affects both temperature and humidity levels.
- ~~**Density of Water -** The density of ocean water is influenced by its temperature and salinity. Areas with higher salinity typically have warmer waters, while lower salinity regions are cooler. This density variation affects ocean circulation and temperature distribution.~~
- **Ocean Currents -** Ocean currents play a crucial role in regulating surface temperatures. Warm currents raise water temperatures and increase evaporation rates, leading to higher rainfall in those regions. Conversely, cold currents can lower temperatures and reduce moisture levels.
- **Local Factors -** Local conditions such as submarine ridges, weather patterns (storms, cyclones), wind, fog, cloudiness, and the rates of evaporation and precipitation all influence ocean temperatures in specific areas.

Vertical Distribution of Ocean Temperature:

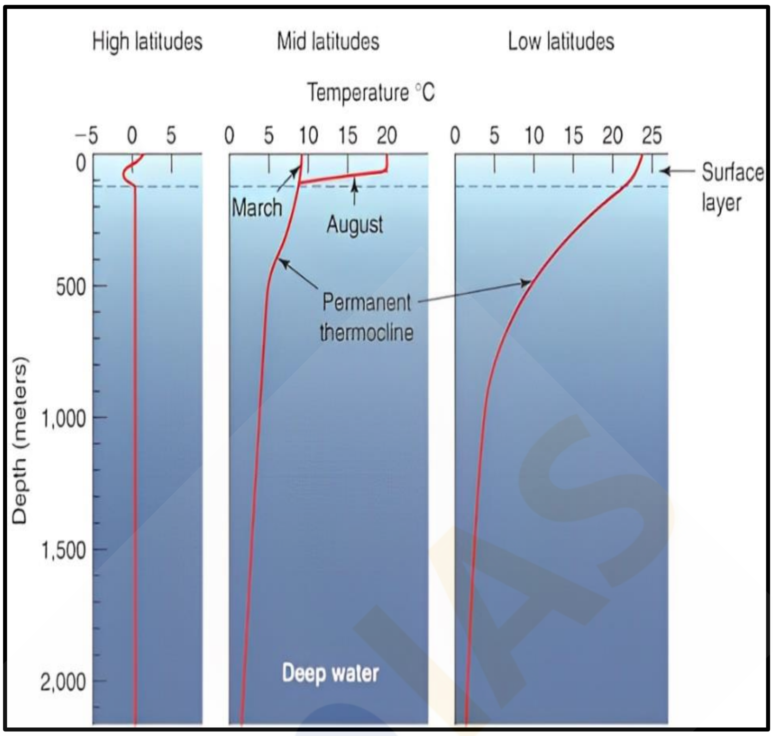
The vertical distribution of temperature in the ocean reveals a **pattern where temperature decreases with increasing depth.** This **decrease is not uniform** but is marked by a **significant boundary region** between the surface waters and the deeper layers of the ocean. This **boundary typically begins at a depth of 100 to 400 meters below the sea surface and extends several hundred meters downward.** This region, known as the **'thermocline,'** is characterized by a **rapid drop in temperature.** Below the thermocline lies about **90% of the ocean's total volume,** where temperatures approach 0°C.

Based on temperature variation with depth, the ocean can be divided into three distinct layers:

- **The Surface Layer -** Also referred to as the **Photic layer or Euphotic layer,** this is the uppermost layer of the ocean, about **500 meters thick,** has temperatures ranging from **20°C to 25°C.** In tropical regions, this warm layer is present year-round, while in mid-latitudes, it typically develops only during the summer months.
- **The Thermocline Layer -** Situated below the surface layer, comprising about **18% of the ocean's total water volume.** The **thermocline is marked by a rapid decrease in temperature with depth.** This layer extends from **500 to 1,000 meters.**
- **The Deep Layer -** Found **below 1000 meters,** particularly in mid-latitude regions, this zone accounts for approximately **80% of the ocean's total water volume.** Here, the temperature remains nearly constant, typically just **1 or 2 degrees Celsius above the freezing point,** even at the ocean's deepest points. In polar regions, particularly within the Arctic and Antarctic circles, surface water temperatures are near 0°C, leading to only a slight temperature change with depth. In these regions, a single layer of cold water extends from the surface to the ocean floor.



Temperature Profiles and Thermocline Variation Across Latitudes:



Low Latitudes (Tropical Regions):

- In low-latitude tropical regions, the **sea surface is significantly warmer,** leading to a **well-defined and highly pronounced thermocline.** The warm surface waters result in a **steep temperature gradient as depth increases.** Additionally, the tropical regions experience **minimal seasonal variations** in surface temperature, which means that the thermocline remains relatively stable throughout the year with little seasonal change in the temperature profiles.

Mid-Latitudes (Temperate Regions):

- Mid-latitude temperate regions **exhibit more pronounced seasonal fluctuations** in surface temperature compared to both polar and tropical regions. The **surface water temperature can vary by 8-15°C between summer and winter,** creating seasonal differences in the thermocline. During the summer, the surface waters warm up significantly, leading to a more pronounced thermocline. However, in the winter, the thermocline becomes deeper as winter storms churn up the surface waters, creating a deeper mixed layer. Consequently, the **thermocline in mid-latitudes is more pronounced in summer and deeper in winter.**

High Latitudes (Polar Regions):

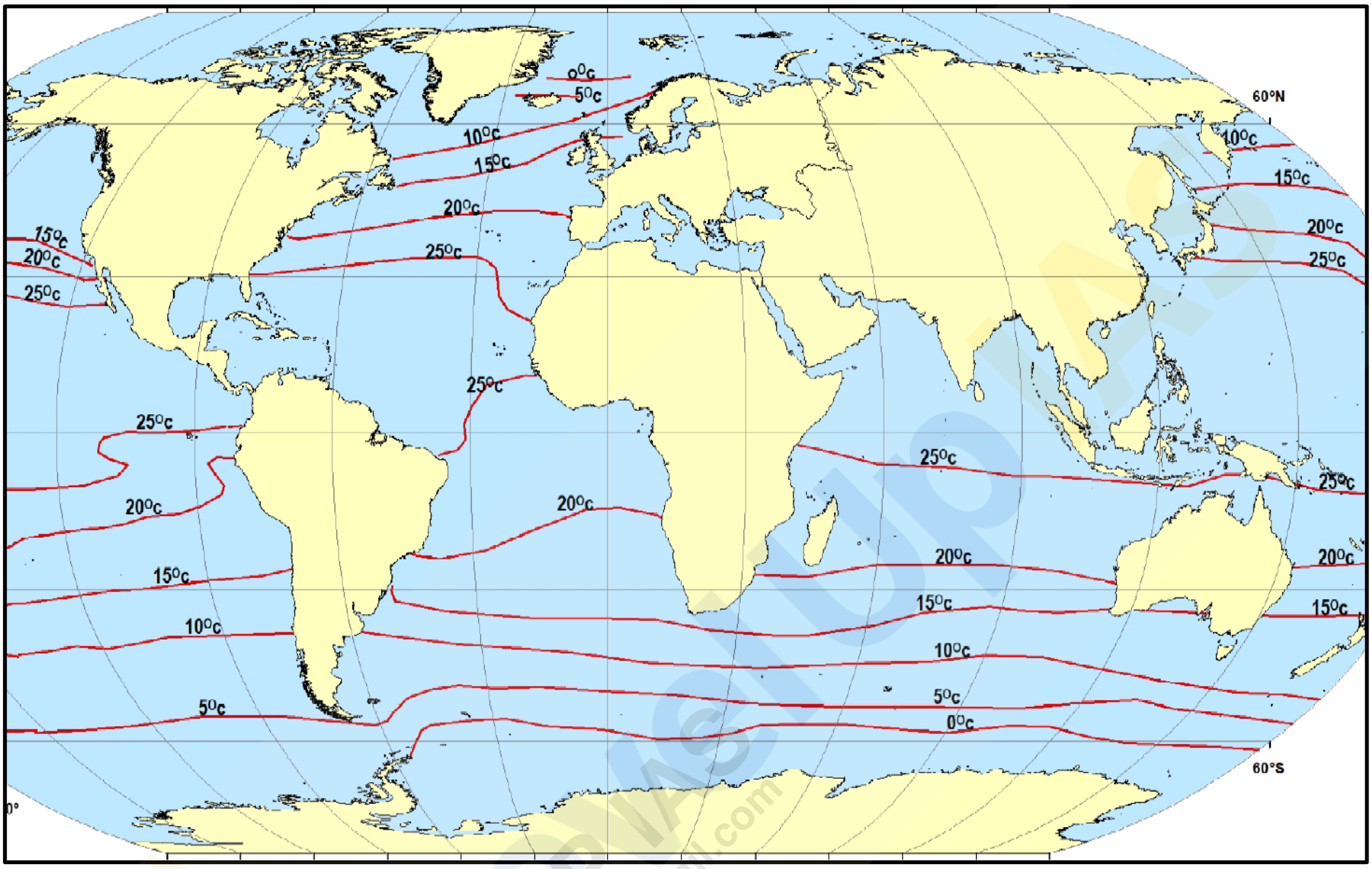
- In high-latitude polar regions, there is **little difference between the surface and deep-water temperatures,** resulting in a **relatively uniform temperature profile.** The water remains cold at all depths, leading to the absence of a strong thermocline. As in tropical regions, polar waters experience **minimal seasonal temperature changes,** with surface temperatures remaining consistently low throughout the year.

Horizontal Distribution of Ocean Temperature:

- The **average temperature of the surface waters of the world's oceans** is approximately **27°C.** This temperature tends to **decrease gradually** as one moves **from the equator towards the poles.** The decline in temperature with increasing latitude generally follows a **rate of about 0.5°C per degree of latitude.** This gradual cooling is a key characteristic of the horizontal distribution of ocean temperatures.
- The horizontal temperature distribution in the oceans is represented by **'Isothermal lines' - lines that connect points of equal temperature.** The spacing of these isotherms varies depending on the temperature gradient. In regions where the temperature difference is significant, isotherms are closely spaced. Conversely, in areas with minimal temperature variation, the isotherms are more widely spaced.
- A notable example of this pattern can be observed in **February,** particularly **south of Newfoundland, near the west coast of Europe,** and in the **North Sea.** In these regions, **isotherms are closely spaced, reflecting a sharp temperature gradient.** However, as these lines extend towards the coast of Norway, they spread out, creating a bulge towards the north. This phenomenon is primarily influenced by the **interaction of ocean currents.** The **cold Labrador Current** flows southward along the North American coast, causing a more pronounced decrease in

temperature compared to other regions at the same latitude. Simultaneously, the **warm Gulf Stream** moves towards the western coast of Europe, **elevating temperatures along the west coast of Europe.** This interplay between cold and warm currents significantly shapes the horizontal temperature distribution in these areas.

- **The maximum ocean temperature** is consistently found at the **surface** because it directly absorbs solar heat, which is then transmitted to deeper layers through convection. However, the **rate at which temperature decreases with depth is not consistent.** The **most rapid temperature decline occurs within the first 200 meters,** after which the rate of decrease slows significantly.



Salinity of Ocean Waters:

All natural waters, including rainwater and ocean water, contain dissolved mineral salts. The term **"salinity"** refers to the **total concentration of these dissolved salts in seawater.** Salinity is typically measured as the **amount of salt** (in grams) **dissolved in 1,000 grams** (1 kilogram) **of seawater** and is expressed in **parts per thousand (ppt or ‰).** An important characteristic of seawater, salinity helps define the nature of marine environments. For instance, **water with a salinity of 24.7‰ is considered the upper threshold for brackish water.**

On average, the **salinity of ocean water is about 35 ppt** at a temperature of zero degrees Celsius, meaning that dissolved salts make up approximately **3.5% of the total weight of ocean water.** Among these dissolved salts, **sodium chloride** (common table salt) is the most abundant.

Sources of Ocean Salinity:

- The salts dissolved in ocean waters **originate primarily from continental landmasses.** These salts are transported to the oceans via rain, rivers, groundwater, sea-waves, winds, and glaciers. Some of the dissolved salts also come from the **ocean floor.** The Earth's layers beneath the crust contain molten minerals, which can reach the surface through volcanic activity or the continuous emission of gases, contributing to the ocean's salinity.

Role of Ocean Salinity:

Salinity plays a crucial role in determining various **physical and chemical properties** of ocean water. It influences the **compressibility, thermal expansion, temperature,** and **density of seawater,** all of which **affect ocean circulation patterns.** Additionally, salinity **controls the absorption of insolation,** impacting the heat balance of the ocean. It also governs the **rate of evaporation** and the **level of humidity in the atmosphere,** particularly over oceanic regions. These factors collectively shape the global climate and weather systems.

Salinity not only affects the physical characteristics of seawater but also has a profound influence on the biological aspects of the ocean. The **movement and distribution of marine species,** including fish and other resources, are closely linked to salinity levels. Variations in salinity can create distinct marine habitats, leading to biodiversity hotspots or areas of low productivity.

Highest salinity in water bodies
Lake Van in Turkey ($330^{\circ}/_{oo}$),
Dead Sea ($238^{\circ}/_{oo}$),
Great Salt Lake ($220^{\circ}/_{oo}$)

Table 13.4 : Dissolved Salts in Sea Water (gm of Salt per kg of Water)

Chlorine	18.97
Sodium	10.47
Sulphate	2.65
Magnesium	1.28
Calcium	0.41
Potassium	0.38
Bicarbonate	0.14
Bromine	0.06
Borate	0.02
Strontium	0.01

Composition of Seawater:

- **Sodium Chloride ($NaCl$) -** 77.7%
- **Magnesium Chloride ($MgCl_2$) -** 10.9%
- **Magnesium Sulphate ($MgSO_4$) -** 4.7%
- **Calcium Sulphate ($CaSO_4$) -** 3.6%
- **Potassium Sulphate (K_2SO_4) -** 2.5%

Factors Affecting Ocean Salinity:

- **Evaporation -** The salinity of ocean surface waters is heavily influenced by the rate of evaporation. In **regions where evaporation is high, salinity tends to increase.** For example, the Mediterranean Sea exhibits higher salinity due to its elevated evaporation rates.
- **Freshwater Influx -** The inflow of freshwater, particularly in coastal regions, significantly affects surface salinity. **Freshwater input from rivers** and the processes of freezing and thawing of ice in polar regions can **reduce salinity levels.** At river mouths, such as those of the Amazon, Congo, and Ganga, the surface salinity of ocean waters is noticeably lower than the average due to the large volumes of freshwater entering the ocean.
- **Temperature and Density - Salinity, temperature, and density of ocean water are closely interconnected.** Any variation in temperature or density will consequently influence the salinity of a given area. Generally, regions with **higher temperatures also exhibit higher salinity levels.**
- **Ocean Currents -** Ocean currents play a crucial role in the spatial distribution of dissolved salts in ocean waters. **Warm currents in equatorial regions tend to push salts away from the eastern margins of oceans,** causing accumulation near the western margins. In temperate regions, ocean currents can increase salinity near the eastern margins, as seen with the **Gulf Stream in the North Atlantic Ocean,** which raises the salinity of waters along the western margins of the Atlantic.
- **Precipitation -** There is an **inverse relationship between precipitation and salinity.** Areas with higher levels of precipitation generally have lower salinity levels. Despite the equatorial region being as warm as the subtropics, it records lower salinity due to the high levels of daily precipitation.
- **Atmospheric Pressure and Wind Direction -** Atmospheric pressure and wind patterns also influence ocean salinity. **Anti-cyclonic conditions, characterized by stable air and high temperatures, can increase the salinity of surface waters.** Winds play a role in redistributing salinity by moving saline waters to areas with lower salinity, thereby decreasing salinity in the former regions and increasing it in the latter.

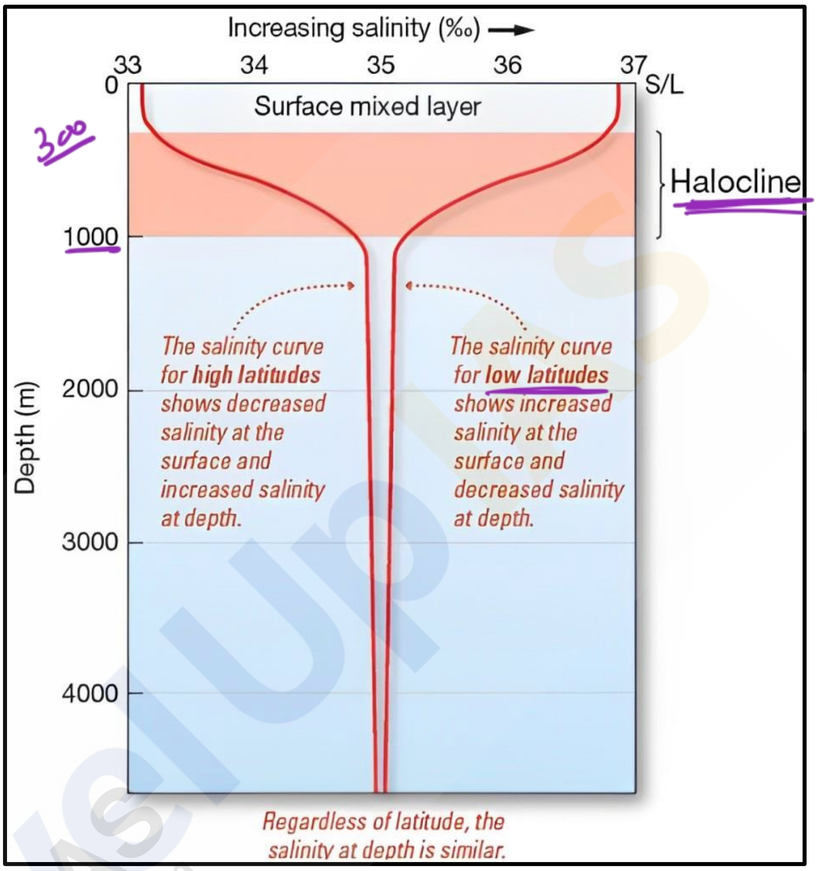
• Ice Melt reduce salinity
• Ice formation increase salinity

Vertical Distribution of Ocean Salinity:

The distribution of salinity within the ocean **varies with depth,** and this variation is influenced by several factors, such as the addition of freshwater and the processes of evaporation and freezing. These processes cause both an increase and decrease in salinity as depth increases, leading to distinct vertical salinity profiles in different latitudes.

Salinity Trends in Different Latitudes:

saline water moves down



High Latitudes: ice formation.

- In high latitudes, **salinity tends to increase with depth between 300 meters and 1,000 meters.** This is because denser, more saline water sinks, creating a positive relationship between depth and salinity. Beyond 1,000 meters, the salinity remains relatively constant.

Low Latitudes: → Evaporation

- In low latitudes, the pattern is somewhat different**. Salinity typically decreases with depth between 300 meters and 1,000 meters.** However, like in high latitudes, beyond 1,000 meters, salinity levels off and becomes nearly constant.

The Halocline:

- The **zone between 300 meters and 1,000 meters** is marked by a **steep gradient in salinity, where rapid changes in salinity occur with depth.** This transitional zone is known as the **'Halocline.'** The halocline serves as a boundary between the upper layer, which exhibits maximum salinity, and the deeper waters, where salinity is lower. In low latitudes, the upper layer of the ocean typically shows higher salinity, while in high latitudes, the salinity is lower.

Horizontal Distribution of Ocean Salinity:

The salinity in open oceans generally ranges between **33‰** and **37‰.** Salinity g**enerally decreases from the equator towards the poles.** However, the **highest salinity levels are not found directly at the equator.** Despite the equator experiencing high temperatures and significant evaporation, the **substantial rainfall in this region dilutes the salinity,** resulting in an average salinity of around **35‰.**

→ Tropics has highest salinity

The highest salinity, approximately 36‰, is typically observed between 20° N and 40° N. This region experiences high temperatures and evaporation but receives relatively low rainfall, contributing to the elevated salinity levels.

In the **southern hemisphere,** between **10° and 30° latitudes,** the average salinity is about **35‰.** Moving towards higher latitudes, between 40° and 60° in both hemispheres, salinity decreases, with averages of 31‰ in the northern hemisphere and 33‰ in the southern hemisphere.

As we approach the **polar regions, salinity continues to decline due to the influx of glacial meltwater.** On average, the northern hemisphere records a salinity of about 35‰, while the southern hemisphere has an average of 34‰.

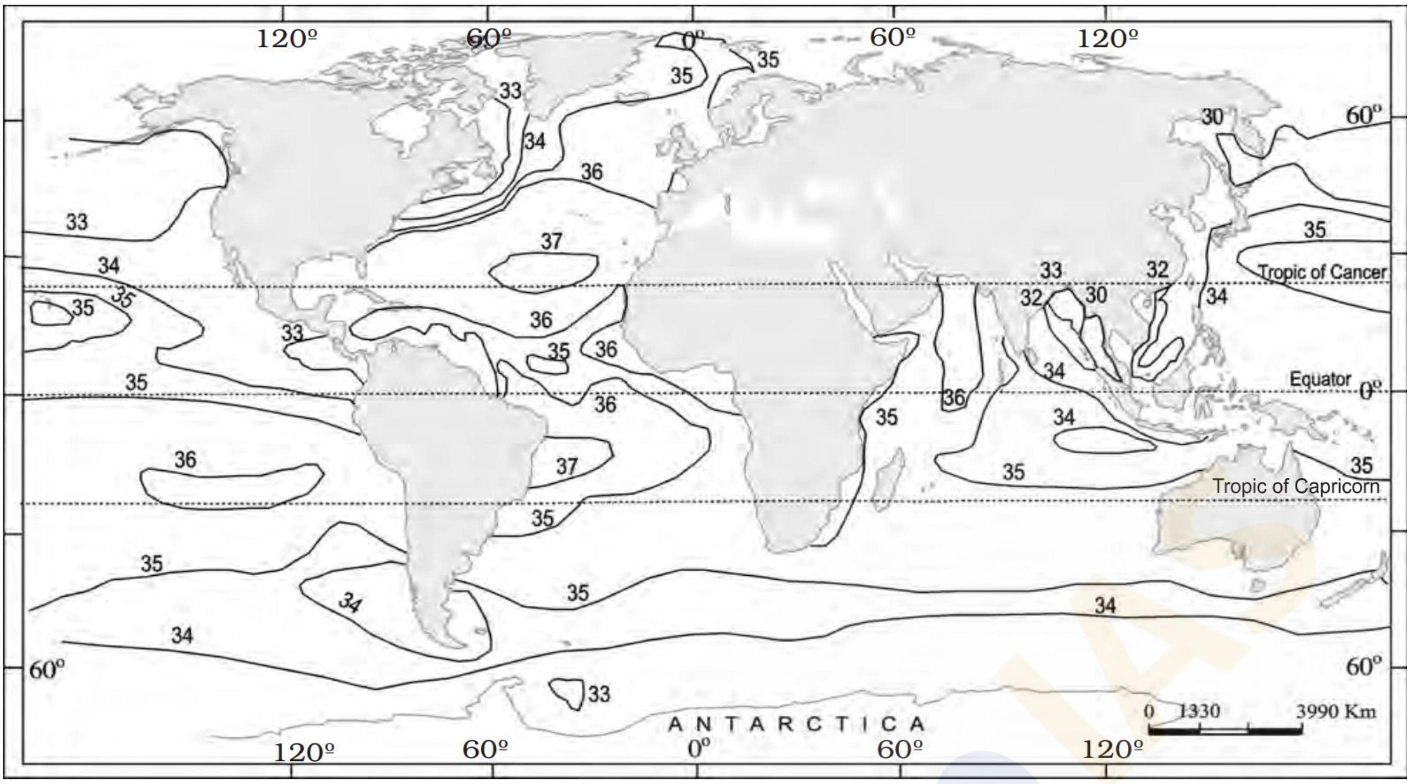


Figure12.5 : Surface salinity of the World's Oceans

Salt Budget (Salt Cycle):

The salt budget, also known as the salt cycle, encompasses the complex processes through which **salt moves between the ocean, lithosphere, and to a minor extent, the atmosphere, before eventually returning to the oceans.**

Salt Movement from Land to Ocean:

- Salt primarily enters the ocean through the **action of moving water, including groundwater.** As water flows over and through rocks, it **leaches minerals through surface erosion.** These minerals, dissolved in water, are carried by rivers and streams that ultimately discharge into the oceans. This continual process contributes to the salinity levels in ocean waters.

Accumulation and Return of Salt to the Lithosphere:

- Within the oceans, some **salts settle to the ocean floor through sedimentation,** forming **mineralized rocks** over time. Over millions of years, geological processes like **plate tectonics and volcanic activity** may raise these salt-laden rocks above the ocean surface, returning them to the lithosphere in the form of minerals.

Salt Transfer to the Atmosphere:

- A small amount of salt from the oceans is **transferred to the atmosphere through wind action.** This airborne salt **eventually returns to the lithosphere when it mixes with precipitation,** though this represents only a tiny fraction of the total salt movement between land and sea.

Duration of the Salt Cycle:

- The salt cycle is a process that operates over an extended geological timeframe, spanning millions of years. **Annually, approximately 3 billion tons of salt are added to the oceans from the land.** A small portion of this salt is extracted by humans for daily use, but the majority continues to circulate within the natural salt cycle.

Regional Distribution of Water Salinity Across the Oceans:

Indian Ocean:

- The Indian Ocean has an **average salinity of 35 parts per thousand.** However, regional variations are observed. For instance, the **Bay of Bengal exhibits lower salinity levels due to the substantial influx of freshwater** from the Ganga River. In contrast, the **Arabian Sea experiences higher salinity levels, primarily driven by high evaporation** rates coupled with a limited influx of freshwater.

Pacific Ocean:

- The salinity in the **Pacific Ocean varies widely, influenced mainly by its vast size and shape.** This ocean's vast area leads to significant regional differences in salinity, shaped by varying climatic conditions and geographical features.

Atlantic Ocean:

- Salinity in the Atlantic Ocean ranges from **20 to 37 parts per thousand,** depending on the location. **Near the equator, factors such as heavy rainfall, high relative humidity, cloudiness, and the calm air of the doldrums contribute to lower salinity levels.** In the **polar regions, minimal evaporation and the significant influx of freshwater from melting ice result in salinity levels ranging between 20 and 32 parts per thousand.**

North Sea:

- Despite its location in higher latitudes, the North Sea records **higher salinity levels.** This is attributed to the **more saline waters carried into the region by the North Atlantic Drift,** which raises the overall salinity.

Mediterranean Sea:

- The Mediterranean Sea is characterized by **higher salinity levels,** with surface waters averaging **about 38 parts per thousand.** This is largely due to the **high rates of evaporation** that prevail in the region, which concentrate the salt content in the water.

Baltic Sea:

- The Baltic Sea, in contrast, records low salinity levels, averaging around **35 parts per thousand.** The primary factor contributing to this **lower salinity is the significant influx of freshwater from numerous rivers** that flow into the sea.

Black Sea:

- The Black Sea has one of the lowest salinity levels among the seas mentioned, ranging between **13 and 23 parts per thousand.** This is due to the **large volumes of freshwater that enter the sea** from surrounding rivers, diluting the salt content significantly.

Red Sea:

- The Red Sea is renowned for its **high salinity.** With **limited freshwater inflow** and **intense evaporation** due to its arid climate, the Red Sea's salt concentration is significantly higher than the global average. Its salinity ranges from between **~36 ‰ in the southern part** and **41 ‰ in the northern part** around the Gulf of Suez, with an **average** of **40 ‰.**

Density of Ocean Waters:

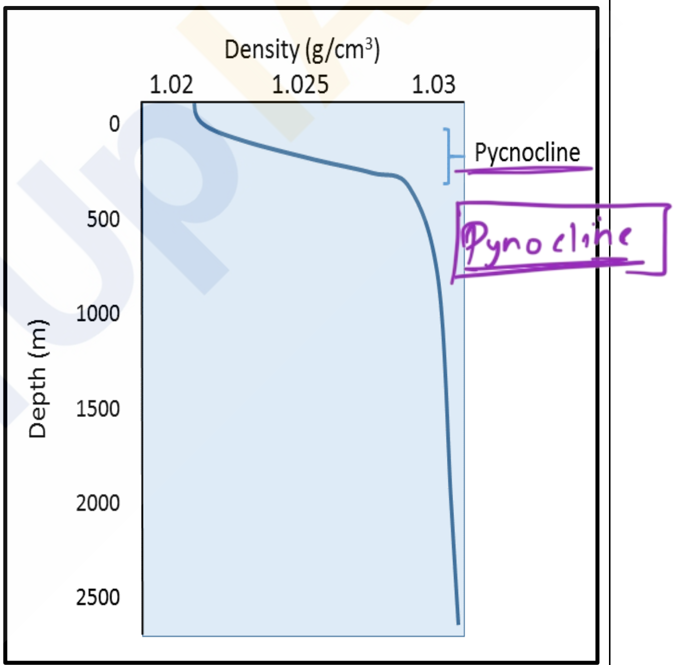
Density refers to the **mass per unit volume of a substance,** commonly expressed in grams per cubic centimeter (g/cm³). **Freshwater** has a **maximum density of 1 g/cm³ at 4°C,** but when salts and other dissolved substances are added, the density of surface seawater increases to a range between 1.02 and 1.03 g/cm³.

Factors Affecting Ocean Density:

- **Temperature -** Temperature is the most significant factor affecting seawater density. As **temperature decreases, the density of water increases.** This is why density profiles of the ocean often mirror temperature profiles.
- **Salinity -** The amount of dissolved salts in ocean water also affects its density. **Higher salinity results in higher density,** as the addition of salts increases the mass of the water without significantly changing its volume.
- **Pressure -** Pressure has the least impact on water density due to **water's incompressibility.** However, at **extreme ocean depths, the slight compression from high pressure slightly increases the density of ocean water.**

Vertical Distribution of Density:

- In the ocean, density typically decreases at the surface where the water is warmest. As depth increases, a region known as the **pycnocline** appears, where there is **a rapid increase in density.** The **pycnocline coincides with the thermocline**, a layer where temperature sharply decreases with depth, leading to an increase in density.
- **Below the pycnocline, density may either stabilize or continue to increase slightly towards the ocean floor.** The stable arrangement of warmer, less dense water above colder, denser water creates a stratified water column. This **stratification prevents mixing between surface and deep waters, forming a barrier to nutrient exchange.**



Thermohaline Circulation:

When denser water forms at the surface due to **cooling or increased salinity,** it becomes **unstable and sinks, displacing less dense water below.** This vertical movement of water masses is known as **thermohaline circulation**, driven by **differences in temperature and salinity.**

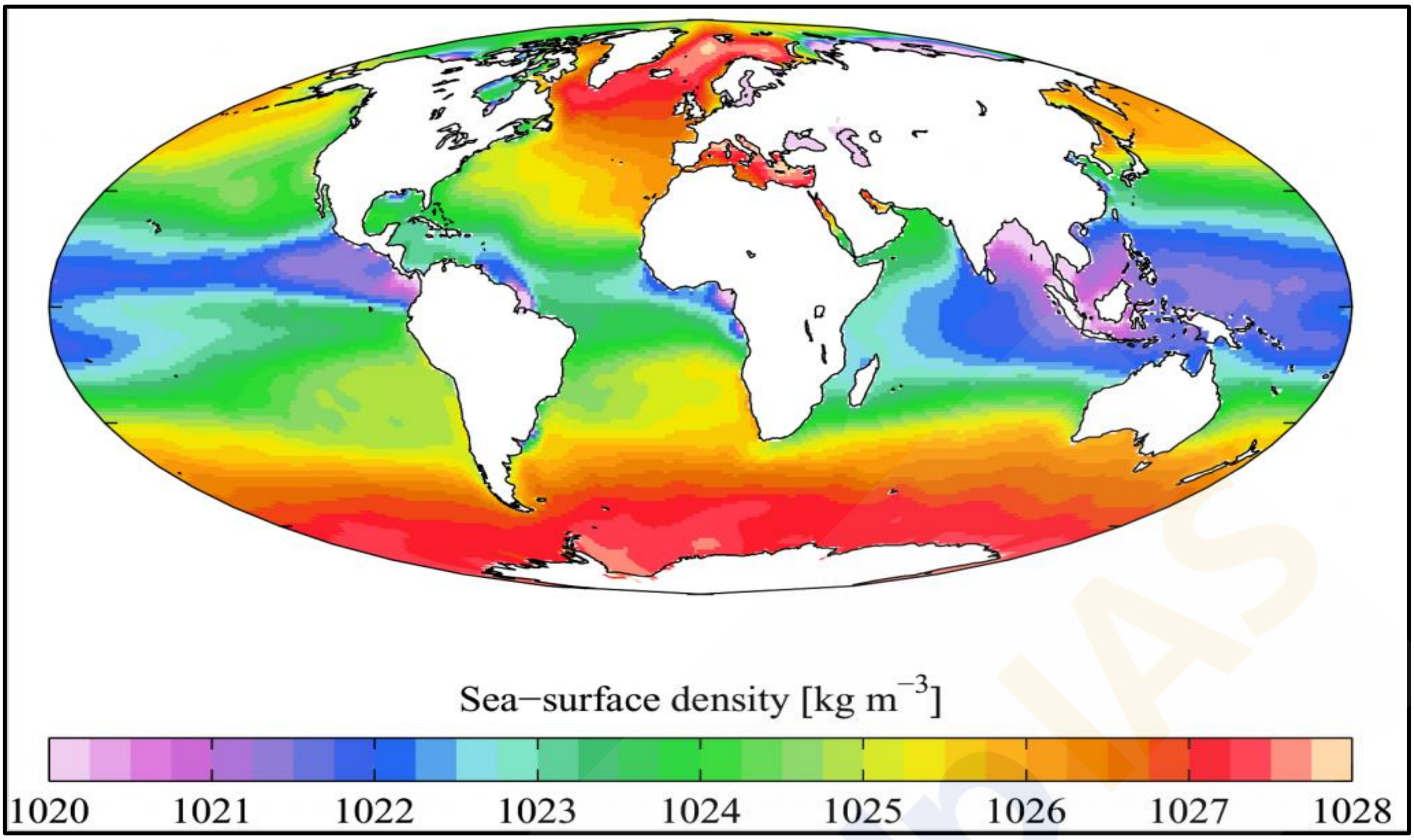
Horizontal Distribution of Density:

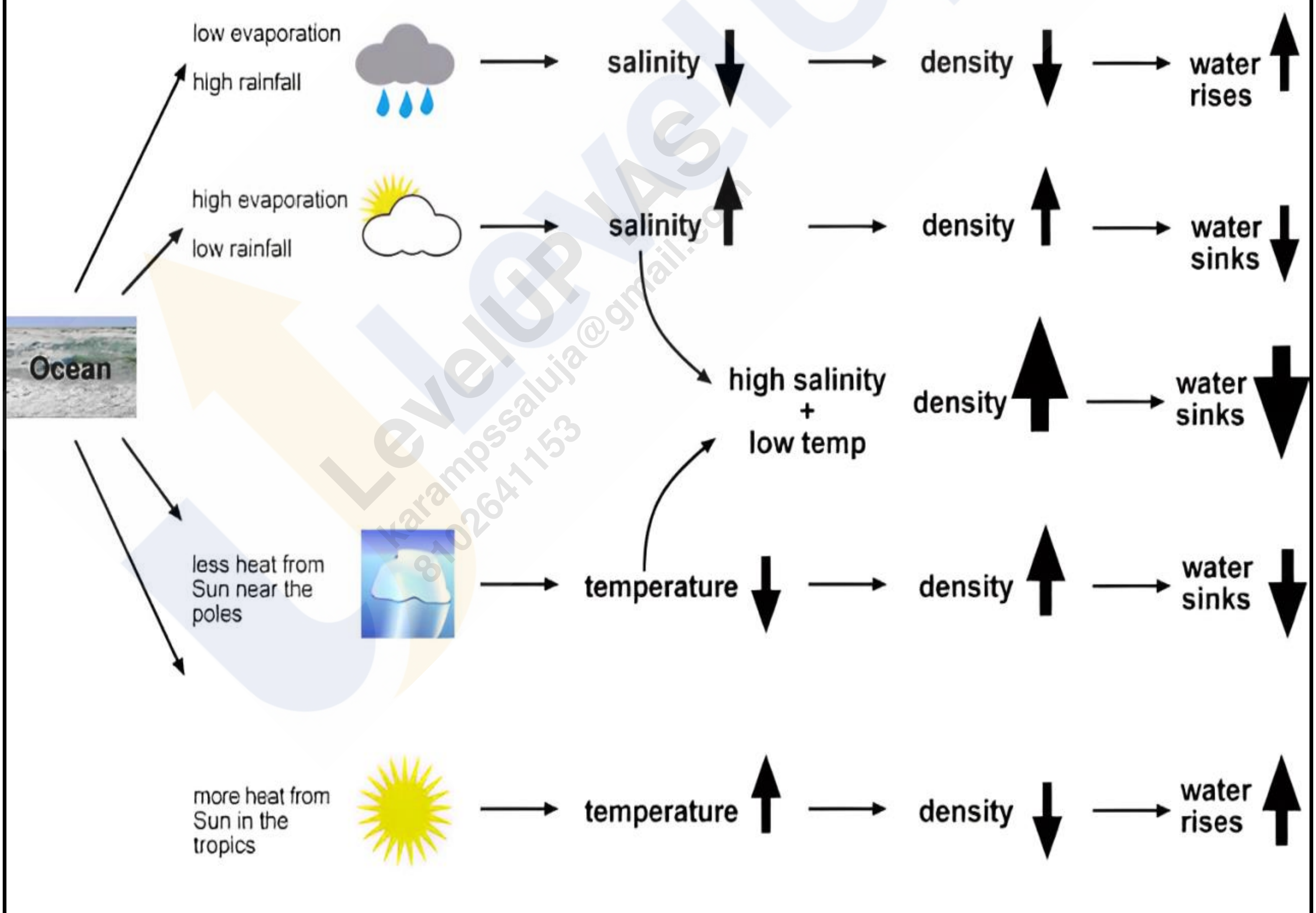
Low Latitudes:

- In the tropics, **warm surface water has low density,** with a **pronounced thermocline separating it from colder, denser deep water.** This strong stratification hinders the upward movement of nutrient-rich water, leading to lower productivity in these regions.

High Latitudes:

- In high latitudes, **the water is uniformly cold across all depths, resulting in minimal density stratification.** The absence of a pycnocline **allows easier mixing between deep and surface waters,** enhancing nutrient availability and supporting higher primary production.


Sea−surface density [kg m^{-3}]
1020 1021 1022 1023 1024 1025 1026 1027 1028


Ocean
low evaporation
high rainfall
salinity
density
water rises
high evaporation
low rainfall
salinity
density
water sinks
high salinity + low temp
density
water sinks
less heat from Sun near the poles
temperature
density
water sinks
more heat from Sun in the tropics
temperature
density
water rises

===== XXXXX =====

1. Ocean Bottom Relief
2. Ocean Property → Temp. Density, Salinity
3. Ocean water movement → current, Tidal wave

4. Coral Reef → Mains

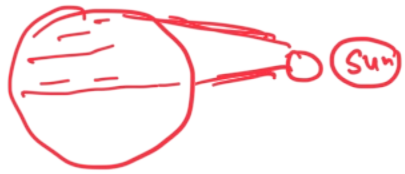
C. Margin
- C. shelf
- C Break
- C slope
- C Rise

Trench
Sea Mount
Guyotz
Vol. Island
MOR
Abbyssal Plain

Ocean
- CC
- OC
  1. Basalt
  2. More denser
  3. Thinner

Temperature

(1) Latitude



Higher latitude ⇒ low temp.

Low latitude ⇒ High temp.
direct sun ray

But Temp at Tropic >> Temp. at Equator

At equator more evaporation & cloud formation
so lacbedo process

(2) Wind →

wind displace warmer water place taken by colder water

up welling → cold water

→ Down welling → hot water

$T_1 > T_2$

GS FOUNDATION 2024-25

GEOGRAPHY - 21

Movements of Ocean Water

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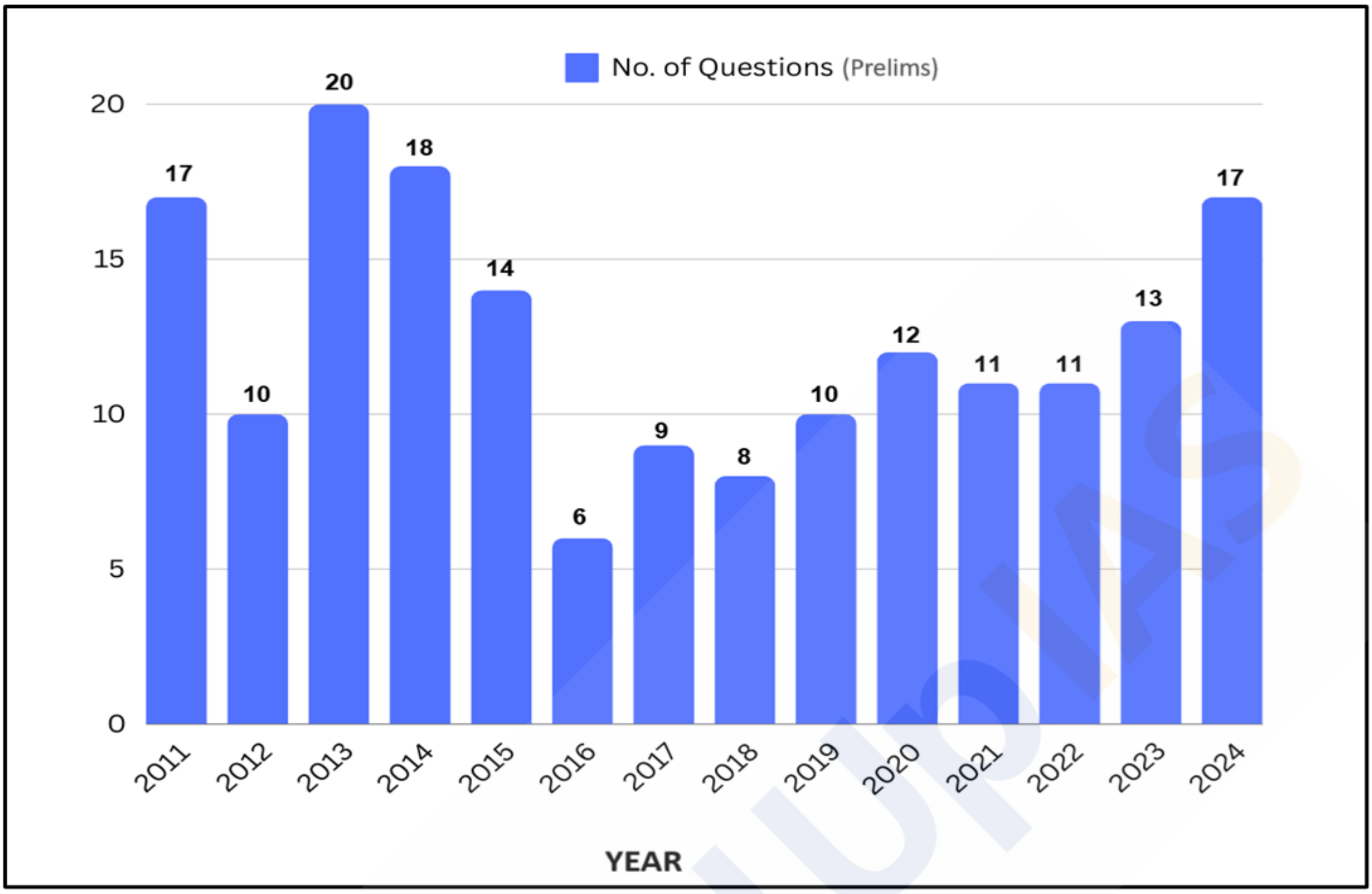
Year-wise Trend Analysis (2011 – 2024)



Topic-wise Trend Analysis (2011 – 2024)

Physical Geography

Topic	Questions
Geomorphology	9
Oceanography	5
Climatology	22
Mapping	16
Distribution of Resources	5
Miscellaneous	15

Indian Geography

Topic	Questions
Physiography of India	24
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Movements of Ocean Water

Introduction:

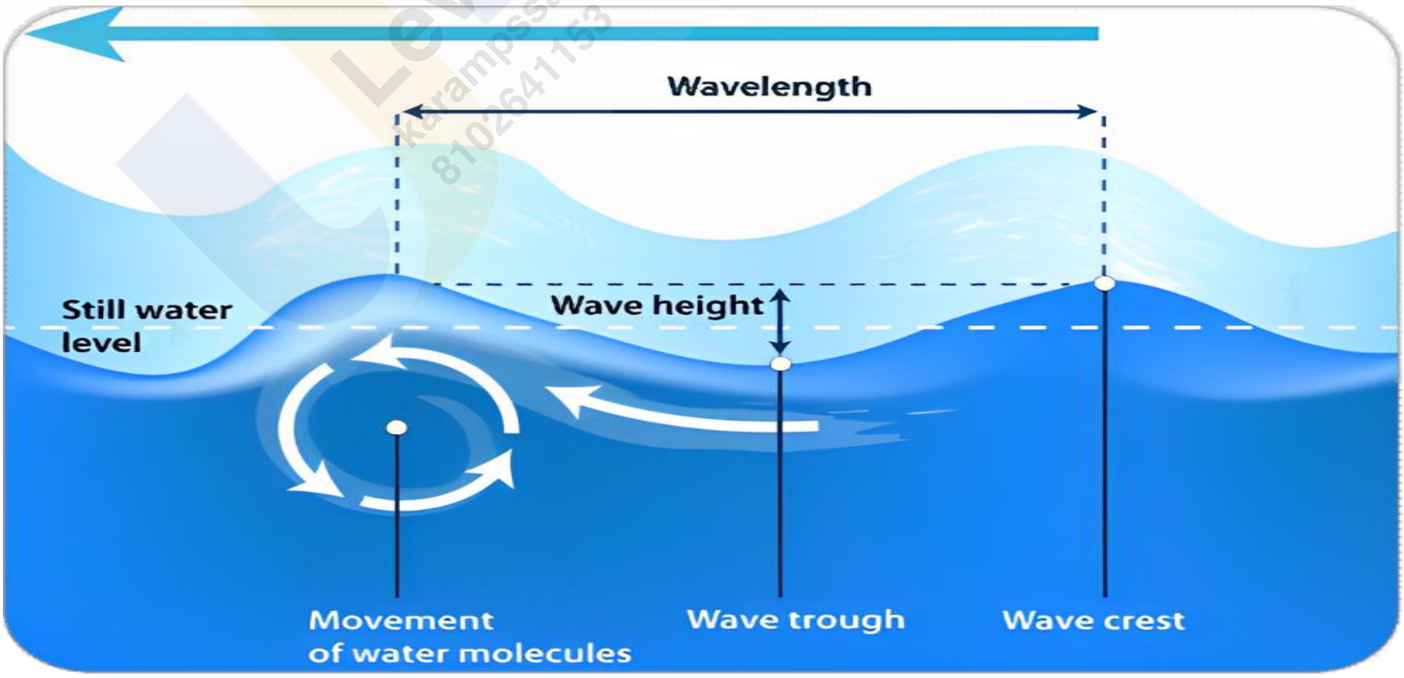
Ocean water is in constant motion, influenced by a variety of physical characteristics and external forces. These movements can be classified into three basic types - **Ocean waves, Tides,** and **Ocean currents.** The temperature, salinity, and density of ocean water, along with the gravitational pull of the sun and moon, and the action of winds, all contribute to these movements. The motions of ocean water can be categorized into **horizontal and vertical movements,** each playing a crucial role in the dynamics of the marine environment.

Ocean Waves:

Ocean waves are **oscillatory surface movements of water,** characterized by the rise and fall of the ocean surface. It represents the **horizontal movement of energy across the ocean's surface,** rather than the physical movement of water from one place to another. **While the energy within a wave moves forward, the water particles themselves only move in small circles.** Waves are primarily generated by **wind,** which **transfers energy to the water.** As waves approach the shore, their speed decreases due to friction with the sea floor, eventually causing them to break. The **largest waves are typically found in open oceans,** where they continue to grow as they absorb more energy from the wind.

Characteristics of Waves:

- **Wave Crest and Trough -** The **crest** is the **highest point** of a wave, while the **trough** is the **lowest point.**
- **Wave Height -** Wave height is the **vertical distance between the bottom of a trough** and the **top of a crest.**
- **Wave Amplitude -** Wave amplitude is **half of the wave height.**
- **Wave Period -** The wave period is the **time interval between two successive crests or troughs** passing a fixed point.
- **Wavelength -** Wavelength is the **horizontal distance between two successive crests.**
- **Wave Speed -** Wave speed refers to the **rate at which a wave moves through the water,** typically measured in **knots.**
- **Wave Frequency -** Wave frequency is the **number of waves passing a given point within a one-second** time interval.



Formation of Waves:

Waves are generated primarily by the wind. As **wind blows over the ocean's surface, it creates small ripples that grow as wind speed increases, eventually forming waves.** The size and shape of a wave can reveal its origin. **Steep waves are usually young and formed by local winds, while slow and steady waves may have travelled from distant locations, possibly even from another hemisphere.**

The motion of waves involves the interaction between wind and gravity. **Wind pushes the water, creating the wave, while gravity pulls the wave's crests downward.** This interaction causes the wave to move forward, with the water beneath the wave following a **circular motion.** As a wave approaches the shore, its speed decreases due to friction, leading to the wave breaking and the release of energy on the shoreline.

Factors Influencing Oceanic Wave Formation:

Primary Factors:

- **Wind Speed -** The speed of the wind blowing across the water surface is a fundamental factor in wave formation. **Stronger winds** transfer more energy to the water, **generating larger and more powerful waves.**
- **Wind Duration -** The duration for which the wind blows in a consistent direction influences the development of waves. Longer durations allow waves to build up more energy and grow in size.
- **Fetch -** Fetch refers to the **distance over which the wind blows without interruption** across the water's surface. A **longer fetch allows the wind to transfer more energy to the water, resulting in larger waves.** The size and strength of waves are directly proportional to the length of the fetch.

Secondary Factors:

- **Water Dept -** The depth of the water significantly influences wave formation and behavior. **In deeper waters, waves can travel faster and with more energy.** As waves approach shallower waters, their speed decreases, causing them to grow in height and eventually break as they reach the shore.
- **Coriolis Effect -** The Coriolis effect, caused by the Earth's rotation, **influences the direction of wave propagation.** This effect **deflects the path of waves to the right in the Northern Hemisphere** and to the **left in the Southern Hemisphere,** altering their movement and impacting coastal wave patterns.
- **Ocean Currents -** Ocean currents can modify wave characteristics by altering their speed and direction. **When waves interact with opposing currents, their height can increase, leading to more powerful and turbulent conditions.** Conversely, waves moving in the same direction as a current may experience reduced height and energy.
- **Geological Features - Submarine topography,** such as underwater **ridges, trenches, and continental shelves, affects wave behavior.** These features can focus wave energy, causing waves to increase in height, or disperse it, leading to smaller waves. Coastal features like headlands, bays, and reefs also influence wave patterns by refracting or reflecting waves.
- **Atmospheric Pressure -** Changes in atmospheric pressure, **particularly during storms or cyclones, can create large waves known as storm surges.** Low-pressure systems cause the sea surface to rise, generating higher waves and increasing the risk of coastal flooding.

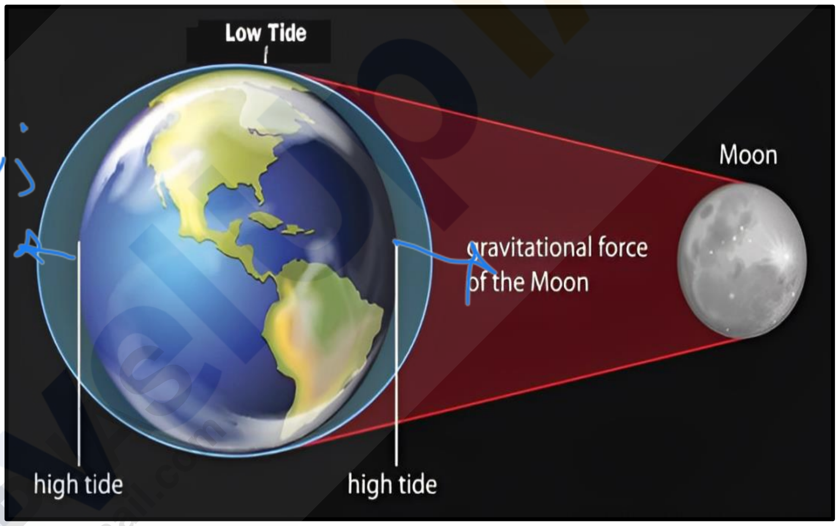
Distribution of Fresh water
Upper → Glacier
Niche → Ground water
L → Lake
A → Atmos
R → River

- **Tides -** Tidal forces, caused by the **gravitational pull of the Moon and the Sun, influence wave formation,** especially in shallow waters. Tides can amplify or dampen wave energy depending on their phase and the interaction with local topography.
- **Earthquakes and Underwater Volcanic Eruptions -** Sudden geological events like **earthquakes or underwater volcanic eruptions can generate seismic sea waves, known as tsunamis.** These waves have immense energy and can travel across entire ocean basins, causing significant impact when they reach coastal areas.

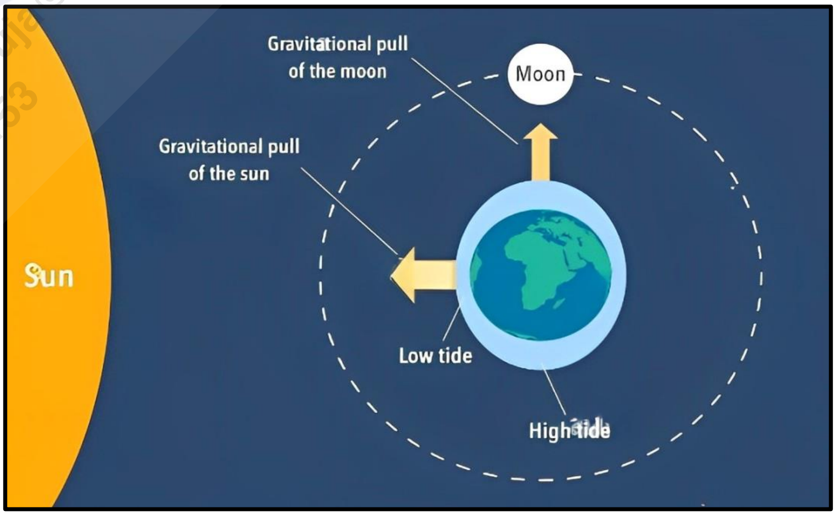
Tides: → vertical movement of surface water

Tides are the **regular rise and fall of sea levels in oceans and seas,** occurring **once or twice daily,** driven by the gravitational forces exerted by the moon and the sun. These natural phenomena are distinct from meteorological effects, such as surges, which are irregular movements of water driven by wind and atmospheric pressure changes. The study of tides is complex due to significant spatial and temporal variations in their frequency, magnitude, and height.

Primary Factors (Forces) Contributing to the Occurrence of Tides:



- **Gravitational Pull of the Moon -** The moon's gravitational force is the **primary driver of tides,** exerting a **significant pull-on Earth's water.**
- **Gravitational Pull of the Sun -** Although the sun is much larger than the moon, its gravitational pull-on Earth's water is less effective due to the greater distance.
- **Centrifugal Force -** This force **acts in opposition to ~~Earth's~~ moons gravitational pull,** contributing to the formation of tides by creating an imbalance in the forces acting on the ocean water.



Other Factors Controlling Tides:

- **Uneven Water Distribution -** The uneven distribution of water across the globe affects tidal patterns.
- **Ocean Configuration -** Irregularities in ocean shape and topography can alter tide behavior.
- **Tidal Currents -** When tides move into bays and estuaries, they form tidal currents, influencing local water movement.

Types of Tides:

Based on Frequency:

Semi-diurnal Tide:

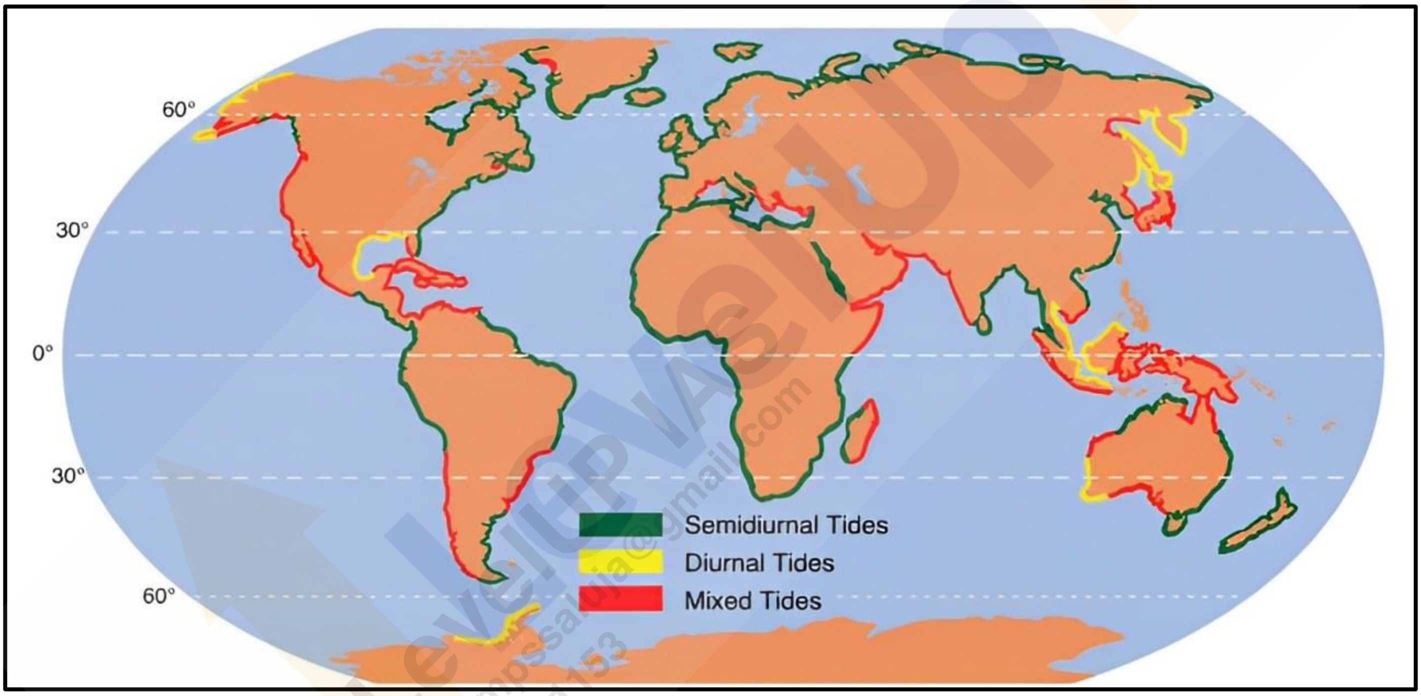
- The most common tidal pattern, featuring **two high tides and two low tides each day,** with successive high or low tides of approximately the same height. Although tides occur twice daily, the interval is not exactly 12 hours but rather **12 hours and 25 minutes.** This **time lag is due to the Moon's west-to-east revolution around the Earth.** An example is **Southampton, England,** which experiences up to **eight tides a day** due to influences from the North Sea and the English Channel.

Diurnal Tide:

- Characterized by **one high tide and one low tide each day,** with successive high and low tides of approximately the same height. This pattern is less common compared to semi-diurnal tides.

Mixed Tide:

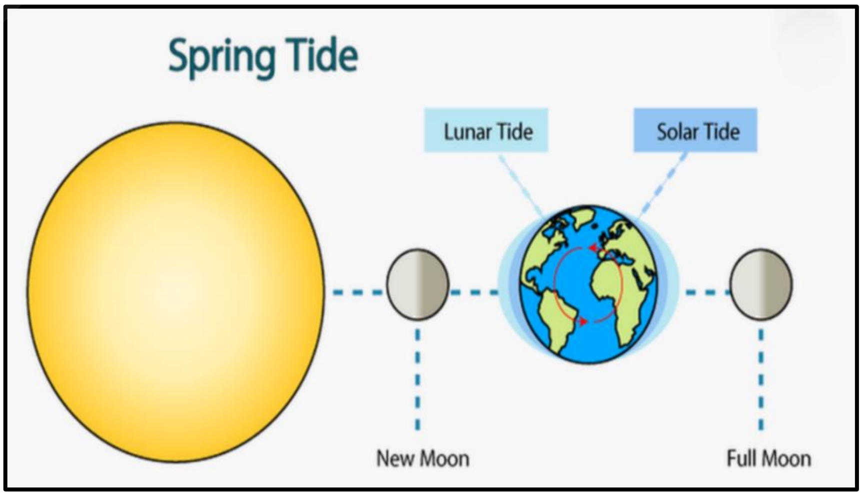
- These tides exhibit **variations in height** and occur along the **west coast of North America** and on many **Pacific Ocean islands.** Mixed tides show a **combination of both diurnal and semi-diurnal characteristics.**



Based on the Sun, Moon, and Earth's Positions:

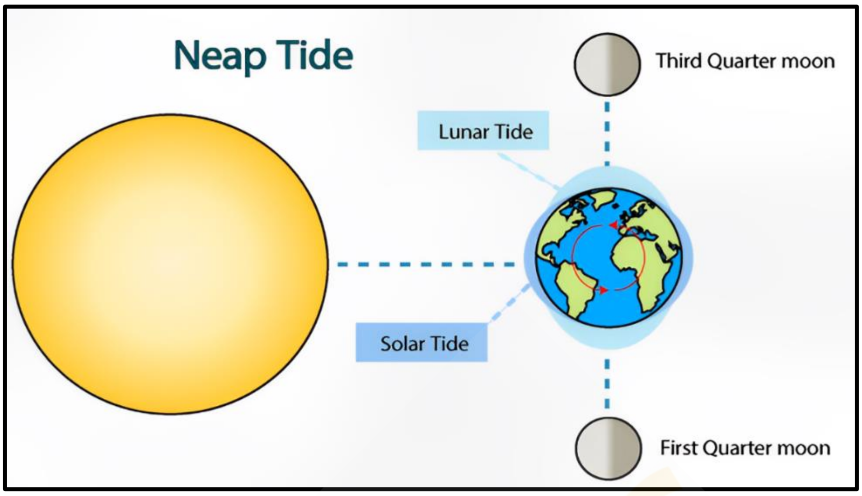
Spring Tides:

- Occur when the **Sun, Moon, and Earth align in a straight line** (a configuration known as syzygy), resulting in **higher-than-normal tides.**
- Spring tides occur **twice a month,** during the **full moon** and new **moon phases.**



Neap Tides:

- Neap tides **occur approximately seven days after spring tides** when the **sun and moon are at right angles to each other.**
- During this time, the **gravitational forces** of the sun and moon partially **counteract each other, reducing the tide's height.**
- Although the moon's attraction is more than twice as strong as the sun's, the sun's gravitational pull diminishes its effect.



Based on Magnitude:

Perigee:

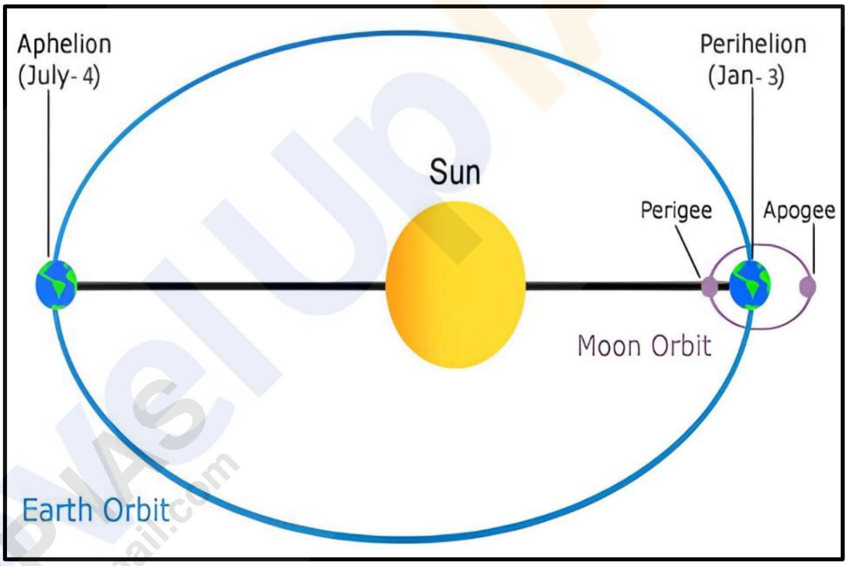
- When the **moon's orbit brings it closest to Earth,** it is at perigee, resulting in unusually **high and low tides.**

Apogee:

- Conversely, when the **moon is farthest from Earth,** it is at apogee, leading to **smaller tidal ranges.**

Perihelion:

- Occurring around **January 3rd,** perihelion is when **Earth is closest to the sun,** causing **unusually high and low tides.**



Aphelion:

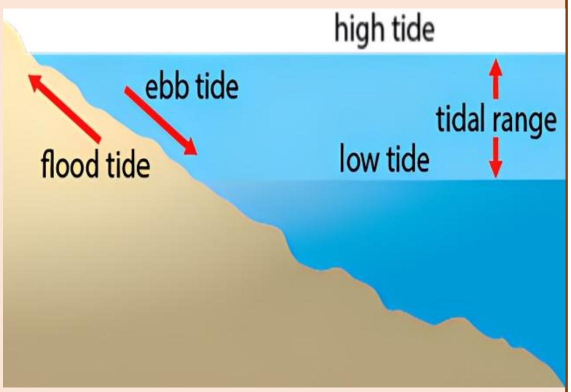
- Around **July 4th,** aphelion occurs when **Earth is farthest from the sun,** resulting in **smaller-than-average tidal ranges.**

Ebb and Flow

*The period when the **tide is falling, and water levels decrease**, is known as the **'Ebb'**.*

*In contrast, the time when the **tide is rising, and water levels increase**, is referred to as the **'Flow or Flood.'***

These alternating periods of ebb and flow are crucial for various coastal and marine activities, influencing navigation, port management, and tidal energy generation.



Importance of Tides:

- **Navigation -** High tides allow large ships to safely enter or leave harbours. Ports like Diamond Harbour in West Bengal and Kandla port in Gujarat utilize tides for safe navigation.
- **River Ports -** Tides assist ships in traveling up river mouths by increasing water volume during high tides, facilitating safe navigation to river ports such as Hooghly (Kolkata), London, and New York.
- **Silt Removal -** Tides help remove silt and sediment from river mouths, keeping them clear for navigation.
- **Prevention of Freezing -** In colder regions, tides prevent the freezing of harbours by bringing in warmer seawater during winter months.
- **Tidal Energy -** Tides are harnessed as a renewable energy source, offering a sustainable means of generating electricity.
- **Fishing -** Tides bring in large volumes of fish, creating productive fishing zones near coastal areas without the need for venturing deep into the sea.

Ocean Currents:

5 major gyres → 4 major current
(1) Pole Boundary current
(2) West Boundary cu.
(3) East Boundary cu.
(4) Equitorial current

Ocean currents are **large, continuous movements of seawater** that flow in a specific direction. These currents are like **rivers within the ocean,** transporting vast volumes of water along defined paths. They are driven by a combination of forces and are essential for distributing heat and nutrients around the globe.

Factors Influencing Ocean Currents:

NH → Clockwise
SH → Anti clockwise

Temp ↑ Density ↓ → Float, rise above

Primary Factors:

- **Solar Heating -** Solar energy causes ocean **water to expand.** As a result, **ocean water near the equator is about 8 cm higher** than in the middle latitudes. This creates a **gradient that causes water to flow downhill.**
- **Wind -** Winds blowing across the ocean surface **exert friction, pushing the water to move in the wind's direction.** Winds **determine the magnitude and direction** of ocean currents, with the Coriolis force also affecting their direction.

  N-E wind lead to Anti-clockwise circulation.

  - **Example -** The **monsoon winds** lead to the **seasonal reversal of currents** in the Indian Ocean. The oceanic circulation pattern often mirrors the Earth's atmospheric circulation, ~~with anticyclonic circulation at middle latitudes and cyclonic circulation at higher latitudes.~~

  For south west monsoon wind lead to clockwise circulation of Water
- **Gravity -** Gravity **pulls water down from higher to lower areas,** contributing to the movement of water along the surface gradient.
- **Coriolis Force -** The **rotation of the Earth affects the movement of water,** causing currents to deflect to the **right in the Northern Hemisphere** and to the **left in the Southern Hemisphere.** This deflection leads to the formation of large circular currents known as **'gyres,'** which dominate the ocean basins.

Secondary Factors:

- **Salinity and Density -** Water density is also **affected by salinity and temperature,** which varies across different regions. **Denser, colder, and saltier water** tends to **sink,** while **lighter, warmer water rises.** This process contributes to the formation of deep-water currents. → Extra page see
- **Obstruction by Land - Landmasses can obstruct the flow of ocean currents,** causing them to **split and change direction.**
  - **Example - The South Equatorial Current** in the Atlantic Ocean is **interrupted by the South American continent,** resulting in the formation of the Brazilian Current, which flows southward in the Atlantic Ocean.

    & CAYENNE CURRENT

Types of Ocean Currents:

Based on Depth:

- **Surface Currents -** These currents **make up about 10% of the ocean's water** and are confined to the **upper 400 meters.** They are **primarily driven by wind** and are responsible for the horizontal movement of water.
- **Deep Water Currents -** Constituting the remaining **90% of ocean water,** these currents are driven by variations in & salinity **density and gravity.** They originate in high latitudes where cold temperatures increase water density, causing it to sink and flow across ocean basins. Salinity ↑ density ↑

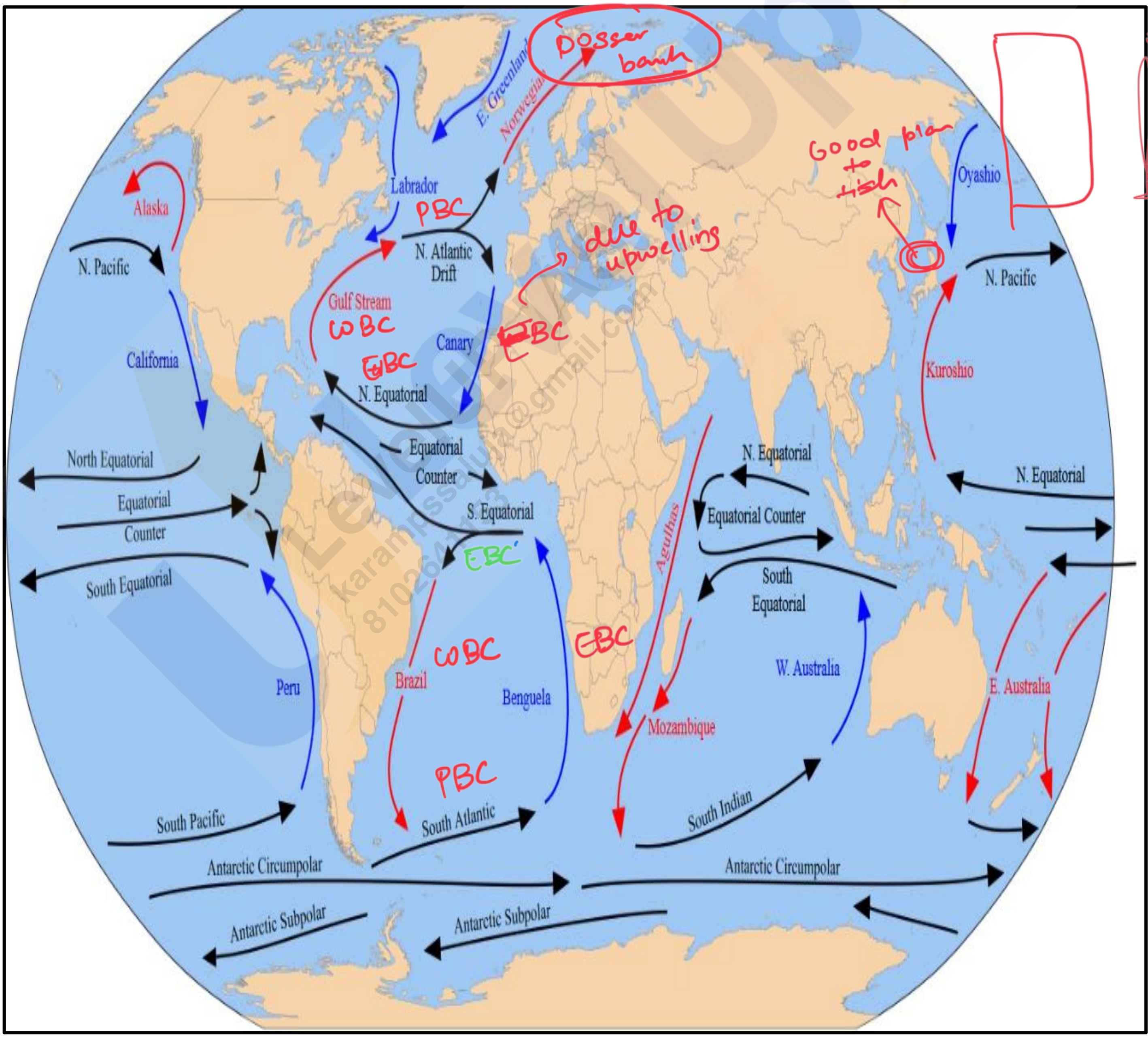
Polar current
Eastern Boundary
Polar Bounda

Based on Temperature:

- **Cold Currents -** These currents carry cold water into warmer regions. They are typically found on the **west coasts of continents in low and middle latitudes** in both hemispheres, as well as on the **east coasts in higher latitudes** of the Northern Hemisphere.

In general
Equitorial current
Western Boundary current

- **Warm Currents -** These currents bring warm water into colder areas. They are usually located on the **east coasts of continents** in low and middle latitudes and on the **west coasts in high latitudes** of the Northern Hemisphere.

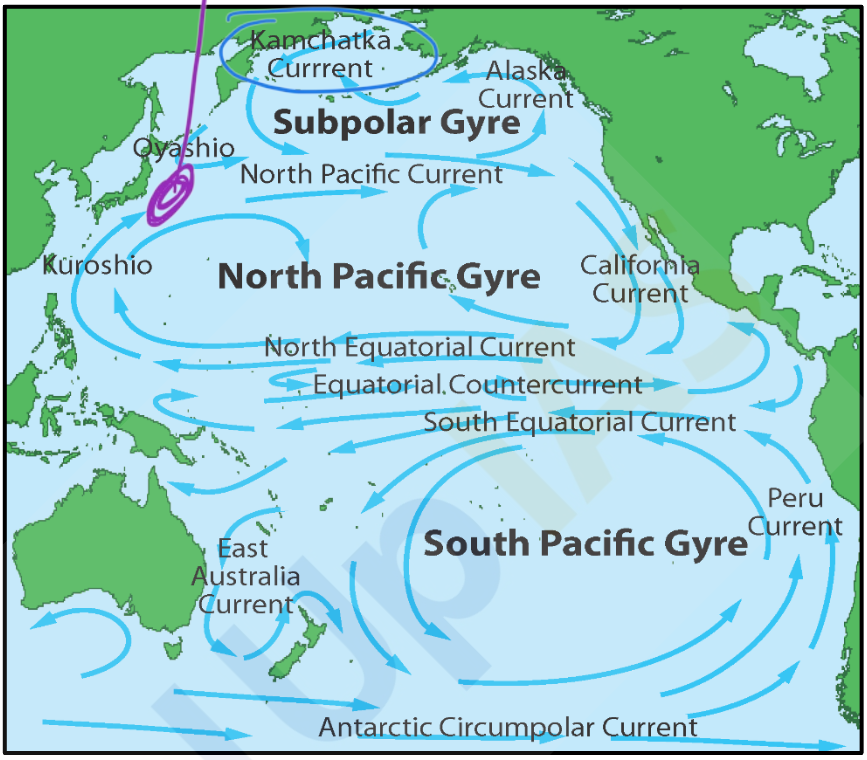


Distribution of Ocean Currents:

{ Good Fishing Ground } in Japan

Pacific Ocean Currents:

The Pacific Ocean, the **largest and deepest ocean** on Earth, features a complex system of currents that play a crucial role in regulating global climate and oceanic ecosystems. These currents are categorized into warm and cold currents based on their temperature and flow characteristics.



Warm Currents:

North Equatorial Current:

- The North Equatorial Current is a warm ocean current that **flows westward across the Pacific Ocean just north of the equator.** It is **driven by the trade winds** and is **part of the larger Pacific Gyre system.** The current originates off the coast of Central America and travels westward toward the Philippines and the eastern coasts of the Asian continent. This current **influences the climate of the western Pacific,** contributing to warmer temperatures and higher humidity levels.

South Equatorial Current:

- The South Equatorial Current **flows westward across the Pacific Ocean just south of the equator.** It is a major warm current **driven by the trade winds,** starting near the coast of South America and extending toward the western Pacific. This current plays a significant role in the distribution of warm water across the southern hemisphere, **affecting weather patterns and marine life** in the region.

Counter Equatorial Current:

- The Counter Equatorial Current, also known as the **Equatorial Counter Current,** is a warm **eastward-flowing current** located **between the North and South Equatorial Currents.** It runs along the equator from the western Pacific towards the eastern Pacific, acting as a **counterbalance to the westward flow of the equatorial currents.** This current helps redistribute warm water across the equatorial Pacific, **influencing the El Niño and La Niña climate phenomena.**

Kuroshio Current System:

- The Kuroshio Current, also known as the **Japan Current,** is a powerful warm ocean current that **flows northward along the east coast of Taiwan and Japan.** It is **part of the North Pacific Gyre** and is analogous to the Gulf Stream in the Atlantic Ocean. The Kuroshio Current is characterized by its **high velocity and warm temperatures,** significantly **impacting the climate of Japan** and the surrounding regions by bringing warm, moist air.

East Australian Current:

- The East Australian Current is a warm ocean current that **flows southward along the east coast of Australia.** It is the largest ocean current in the region and plays a vital role in transporting warm tropical waters toward the temperate zones. The current **influences Australia's eastern coastal climate, promoting a mild and humid environment.** It also **supports a diverse range of marine life** and is popularized as the current featured in the animated film ***'Finding Nemo.'***

North Pacific Drift:

- The North Pacific Drift, also known as the **North Pacific Current,** is a broad, **slow-moving warm ocean current** that **flows eastward** across the northern Pacific Ocean. It forms the northern **boundary of the North Pacific Gyre, originating from the Kuroshio Current.** As it moves eastward, it **splits into several currents,** including the **Alaska Current** and the **California Current.** This drift contributes to the milder climate of the North American west coast.

Cold Currents:

Oyashio Current:

- The Oyashio Current, also known as the **Kuril Current,** is a cold ocean current that flows southward from the Arctic Ocean, along the **eastern coast of Russia and the Kuril Islands, and down to Japan.** It is a **nutrient-rich current** that **supports a productive marine ecosystem,** contributing to the abundance of fish in the region. The Oyashio Current **meets the warm Kuroshio Current near Japan, creating an area of intense oceanic activity and rich biodiversity.**

California Current:

- The California Current is a cold ocean current that **flows southward along the western coast of North America, from southern British Columbia to the southern tip of Baja California.** It is **part of the North Pacific Gyre** and plays a significant role in **moderating the climate of the western United States.** The current **brings cool, nutrient-rich waters** that support a **diverse marine ecosystem,** including important fisheries. It also influences coastal weather patterns, contributing to the characteristic fog of the California coast.

Peruvian or Humboldt Current:

- The Peruvian Current, commonly known as the **Humboldt Current,** is a cold ocean current that **flows northward along the western coast of South America, from southern Chile to northern Peru.** It is one of the **most productive marine ecosystems in the world, supporting rich fisheries due to the upwelling of cold, nutrient-rich waters.** The Humboldt Current plays a critical role in regulating the climate of western South America, **contributing to the arid conditions of the Atacama Desert.**

Atlantic Ocean Currents:

The Atlantic Ocean is home to several major ocean currents, which play a critical role in regulating the climate and weather patterns across the globe. These currents are broadly classified into warm and cold currents, based on the temperature of the water they carry.

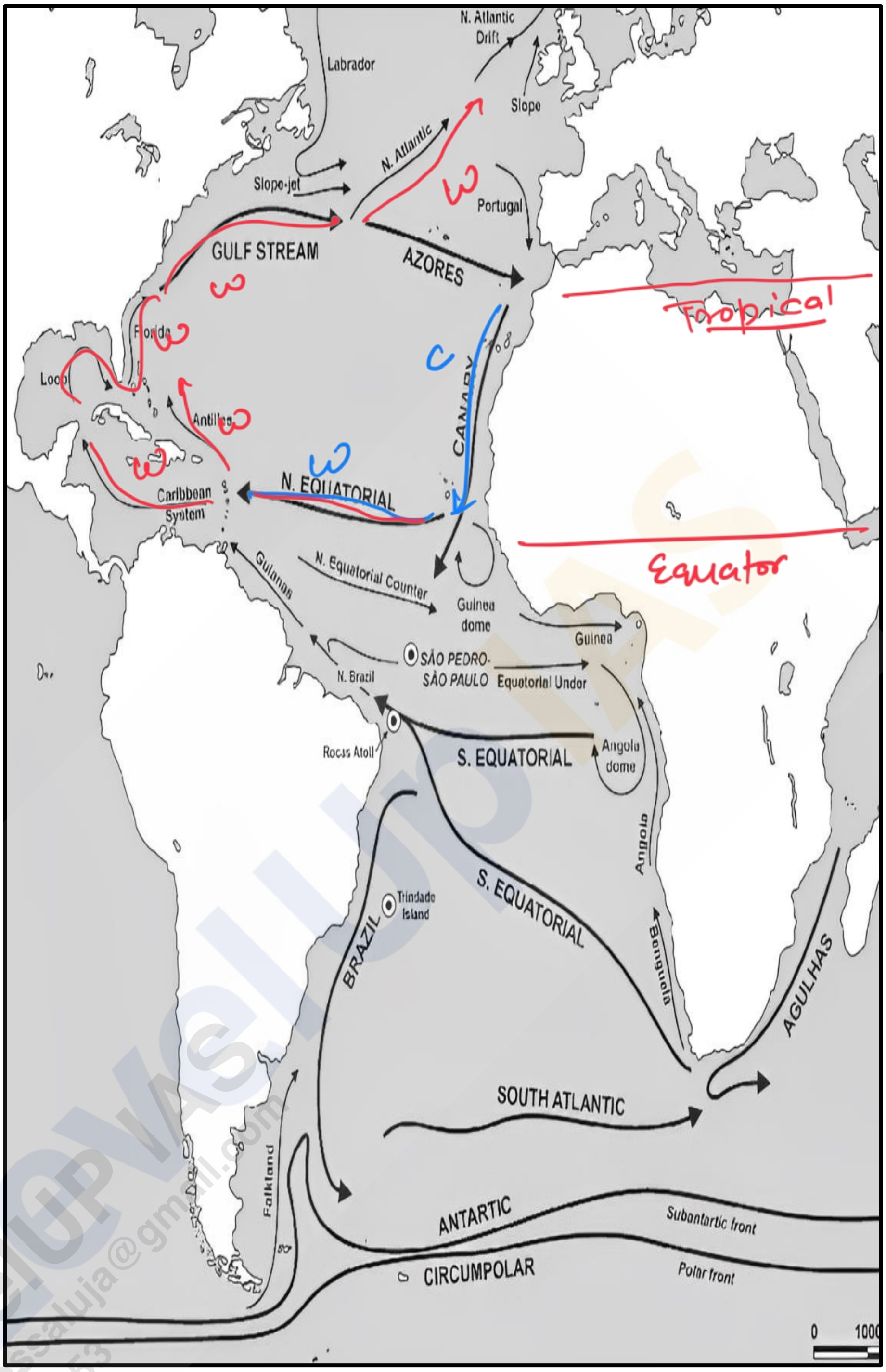
Warm Currents:

North Equatorial Current:

- The North Equatorial Current is a major warm current in the Atlantic Ocean that **flows westward between the equator and about 10°N latitude.** It is primarily **driven by the trade winds** and transports warm water from the eastern Atlantic, near the **African coast, toward the western Atlantic, near the Caribbean and the Americas.** This current plays a significant role in the **heat transfer across the equator,** contributing to the overall thermohaline circulation.

South Equatorial Current:

- The South Equatorial Current is the southern **counterpart to the North Equatorial Current.** It **flows westward between the equator and about 20°S latitude,** moving warm water from the eastern Atlantic, near Africa, toward the northeastern coast of South America. The South Equatorial Current **splits upon reaching the South American coast,** with part of it feeding into the **Brazil Current,** a prominent western boundary current.



Equatorial Counter Current:

- The Equatorial Counter Current is an **eastward-flowing warm current located between the North and South Equatorial Currents.** It is found near the equator, typically between **3°N and 10°N.** This current forms as a result of the accumulation of water in the western Atlantic due to the westward-flowing North and South Equatorial Currents. The water **flows back eastward along the equator,** forming the Equatorial Counter Current, which helps balance the water distribution across the Atlantic.

Gulf Stream:

- The Gulf Stream is one of the most significant warm currents in the Atlantic Ocean, known for its powerful flow and influence on climate. **Originating in the Gulf of Mexico, it flows northward along the eastern coast of the United States** before turning eastward toward Europe. The Gulf Stream **carries warm, tropical water from the Caribbean Sea and the Gulf of Mexico across the North Atlantic, significantly impacting the climate of the eastern United States and Western Europe.** It is also a crucial component of the Atlantic Meridional Overturning Circulation (AMOC), contributing to global thermohaline circulation.

deep current movem

Florida Current:

- The Florida Current is a warm current that **flows northward between the Straits of Florida and the Bahamas.** It is a **continuation of the Gulf Stream** and forms as the waters of the Gulf Stream exit the Gulf of Mexico. The Florida Current is relatively **narrow but has a high velocity,** transporting warm water along the southeastern coast of the United States. It **merges with the Gulf Stream as it moves northward.**

Brazilian Current:

- The Brazilian Current is a warm, **western boundary current** that flows southward along the eastern coast of South America, from around **10°S to 30°S latitude.** It is the **southern extension of the South Equatorial Current** and carries warm, tropical water from the equatorial region toward the southern Atlantic. The Brazilian Current **influences the climate of the southeastern coast of Brazil** and plays a role in the **South Atlantic Gyre,** a major system of ocean currents in the South Atlantic Ocean.

o Norwegian current : Warm current. place Norway Port ice free.

Cold Currents:

Canaries Current:

- The Canaries Current is a cold, eastern boundary current that **flows southward along the northwestern coast of Africa.** Originating from the **North Atlantic Drift,** it transports cool water from higher latitudes toward the equator. The Canaries Current plays a **crucial role in the formation of the dry, arid conditions of the Sahara Desert** and the **Canary Islands,** as it cools the air above, reducing the likelihood of precipitation.

Labrador Current:

- The Labrador Current is a cold current that **flows southward along the eastern coast of Canada, from the Arctic Ocean down to the northeastern coast of the United States.** It carries cold, Arctic water and icebergs into the North Atlantic, contributing to the cold climate of the Canadian Maritimes and the northeastern United States. The **Labrador Current meets the Gulf Stream off the coast of Newfoundland,** creating an area of significant fog and unstable weather known as the **Grand Banks.**

Cold Polar current + west Boundary current

Falkland Current:

- The Falkland Current, also known as the **Malvinas Current,** is a cold, Polar ~~western boundary~~ current that flows northward along the **eastern coast of South America,** from the southern tip of the continent to around 30°S latitude. It **originates from the Antarctic Circumpolar Current** and **transports cold, nutrient-rich water northward along the coast of Argentina.** The Falkland Current influences the climate of southern South America and contributes to the rich marine ecosystems found in the region.

South Atlantic Drift:

- The South Atlantic Drift, also known as the **West Wind Drift,** is a cold, **eastward-flowing current that circles the southern Atlantic Ocean.** It forms **part of the Antarctic Circumpolar Current,** which flows continuously around Antarctica. The South Atlantic Drift carries cold water from the southern Atlantic toward the Indian and Pacific Oceans, playing a role in the global circulation of ocean water and influencing the climate of the southern Atlantic and adjacent continents.

Benguela Current:

- The Benguela Current is a cold, eastern boundary current that **flows northward along the southwestern coast of Africa.** It originates from the **South Atlantic Drift and transports cold, nutrient-rich water from the southern Atlantic toward the equator.** The Benguela Current has a significant impact on the climate and marine life along the coast of Namibia and South Africa, **supporting some of the world's most productive fisheries.** It also contributes to the **arid conditions of the Namib Desert** by cooling the air above and reducing rainfall.

cold Air less water hold capacity

Sargasso Sea

- *The Sargasso Sea is a unique region of the* ***North Atlantic Ocean,*** *characterized by its clear blue water and* ***abundance of floating seaweed*** *called* ***'Sargassum.'***
- *It is located between the* ***Gulf Stream*** *to the west, the* ***North Atlantic Drift*** *to the north, the* ***Canary Current*** *to the east, and the* ***North Equatorial Current*** *to the south.*
- *Unlike other seas, it has* ***no land boundaries.*** *It is defined by its ocean currents, creating a distinct ecosystem within the Atlantic Ocean.*
- *It* ***supports a diverse array of marine life,*** *providing habitat and* ***breeding grounds*** *for many species, including eels, turtles, and a variety of fish.*
- *It is known for its* ***calm waters and low nutrient levels,*** *resulting in* ***high water clarity.*** *The region plays a crucial role in the North Atlantic's carbon cycle, as the* ***Sargassum acts as a carbon sink.***

Labr NAD Azores Florida Gulf Str. Canary NEC SEC Benguela

Board

Falkort

SPC

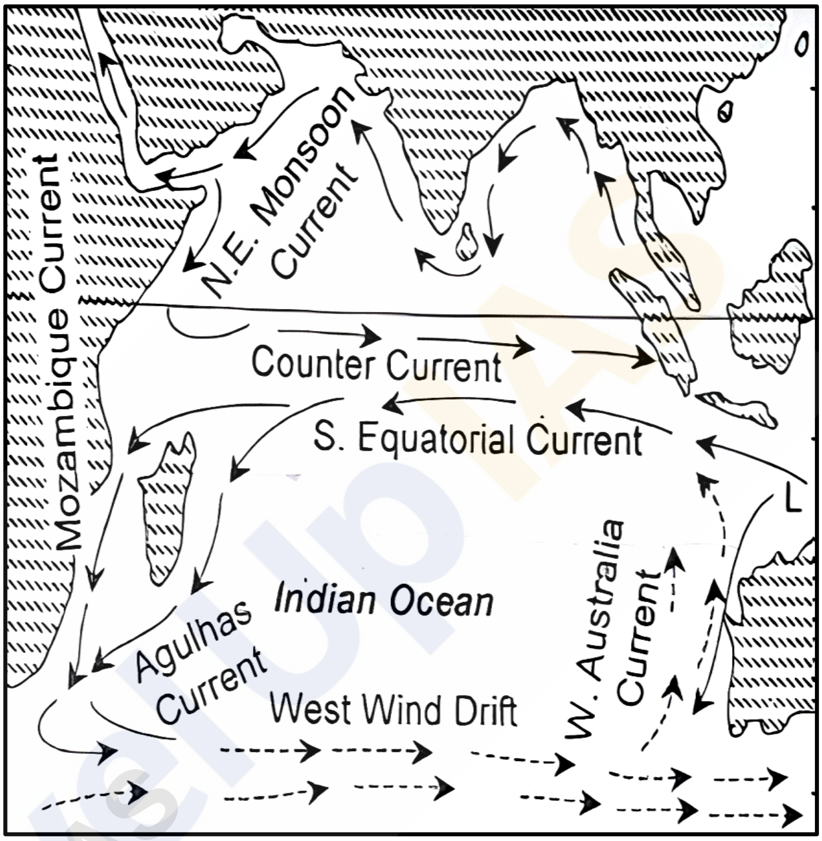
Indian Ocean Currents:

The Indian Ocean, bordered by Asia, Africa, Australia, and the Indian subcontinent, features a **complex system of ocean currents influenced by monsoonal winds, the earth's rotation, and the distribution of landmasses.** The currents in the Indian Ocean can be categorized into drifts, warm currents, and cold currents. Understanding these currents is crucial for comprehending the ocean's impact on climate, weather patterns, and marine biodiversity.

Drifts:

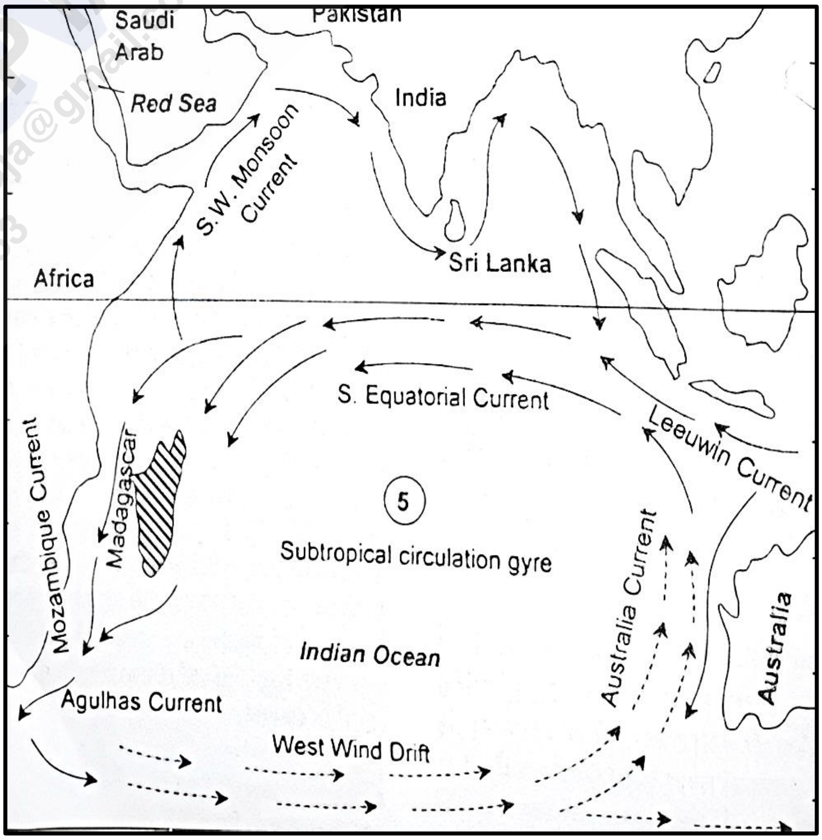
North East Monsoon Drift:

- The North East Monsoon Drift is a **seasonal ocean current** that forms in the **northern part of the Indian Ocean** during the **winter months.** This drift is driven by the Northeast Monsoon winds, which **blow from the northeast, pushing surface waters south-westward.** This movement is typically observed from **November to February,** when this monsoon winds are strongest. The drift's flow is characterized by a broad, relatively slow movement of water, and it plays a crucial role in the regional climate, influencing the weather patterns over the Indian subcontinent and surrounding areas.



South West Monsoon Drift:

- The South West Monsoon Drift occurs during the **summer months** when the **Southwest Monsoon winds dominate the northern Indian Ocean.** This drift is directed **north-eastward, as the monsoon winds push the surface waters towards the Asian landmass.** The South West Monsoon Drift is particularly **strong between June and September,** coinciding with the peak of the Southwest Monsoon. This drift contributes to the **monsoon rainfall over the Indian subcontinent,** significantly affecting agriculture and water resources in the region.



Warm Currents:

North Equatorial Current:

- The North Equatorial Current is a major **westward-flowing** ocean current located just north of the equator in the Indian Ocean. It is **driven by the trade winds** and extends from the eastern coast of Africa to the western shores of Southeast Asia. The North Equatorial Current is a warm current, **transporting warm water across the Indian Ocean.** It plays a significant role in the heat distribution in the region, influencing the climate patterns and marine ecosystems.

South Equatorial Current:

- The South Equatorial Current **flows westward** in the southern part of the Indian Ocean, just south of the equator. This current is **powered by the persistent southeast trade winds** and is a significant warm current, moving large volumes of warm water from the eastern Indian Ocean towards the African coast. The South Equatorial Current is an important component of the global conveyor belt, contributing to the circulation of heat and salinity across the world's oceans.

Mozambique Current:

- The Mozambique Current is a warm ocean current that flows southward along the **eastern coast of Mozambique in Southeast Africa.** It is a **continuation of the South Equatorial Current** and **merges into the Agulhas Current** further south. The Mozambique Current is characterized by its warm temperatures and strong flow, contributing to the overall circulation in the southwestern Indian Ocean. It plays a key role in the regional climate and marine biodiversity, **supporting a variety of marine species along the African coast.**

Madagascar Current:

- The Madagascar Current is a warm ocean current that flows along the **eastern coast of Madagascar.** It is divided into **two branches - the North Madagascar Current,** which flows northward, and the **East Madagascar Current,** which flows southward. The Madagascar Current **originates from the South Equatorial Current** and contributes to the movement of warm water around Madagascar, influencing the local climate and supporting rich marine ecosystems.

Agulhas Current:

- The Agulhas Current is **one of the strongest ocean currents in the world,** flowing southward along the **southeastern coast of Africa.** It is a **continuation of the Mozambique Current and the East Madagascar Current.** The Agulhas Current is a warm and swift current that transports warm water from the tropical Indian Ocean towards the southern tip of Africa, where it retroflects and flows back into the Indian Ocean or mixes with the Atlantic Ocean. This current is crucial for the global ocean circulation, influencing the climate and weather patterns in the region and beyond.

Cold Currents:

Somali Current:

- The Somali Current is a unique ocean current in the western Indian Ocean, **characterized by its reversal in flow direction with the changing monsoon seasons.** During the **Southwest Monsoon (summer), the Somali Current flows northward along the eastern coast of Somalia, carrying cold, nutrient-rich waters from the Arabian Sea.** This cold current supports one of the most productive fishing grounds in the Indian Ocean due to the upwelling of deep, nutrient-rich waters. **During the Northeast Monsoon (winter), the current weakens and reverses direction, flowing southward as a warm current.**

West Australian Current:

- The West Australian Current, also known as the **Leeuwin Current,** is a cold ocean current that **flows northward along the western coast of Australia.** It originates from the Southern Ocean and carries cold, nutrient-poor waters along the Australian coast. The West Australian Current is weaker compared to other major ocean currents and is known for its role in moderating the climate along the western coast of Australia. It influences the marine environment, including the distribution of marine species and the local weather patterns. ∴ help desert for. no less rainfall

Effects of Ocean Currents:

Desert Formation: cold ocean current → cold wind above → less water ↑ retention property

- **Cold ocean currents** play a crucial role in the formation of deserts along the **western coasts of tropical** and **subtropical continents.** These currents contribute to the **desiccating effect, leading to a loss of moisture** and **arid conditions.** The presence of cold currents often results in foggy weather, which further inhibits rainfall and supports desertification in these regions.

Rainfall Patterns:

- **Warm ocean currents** bring rainfall to coastal areas and can even extend their effects to interior regions. For example, the summer rainfall characteristic of the British-type climate is influenced by warm ocean currents. These currents typically **flow parallel to the east coasts of continents in tropical and subtropical latitudes,** creating warm and rainy climates. Such regions often lie on the **western margins of subtropical anticyclones.**

Moderating Temperature:

- Ocean currents have a moderating effect on coastal temperatures. For instance, the **North Atlantic Drift brings warmth to England,** while the **Canary Cold Current has a cooling effect** on countries like **Spain and Portugal.** This moderation helps create milder climates along coastlines compared to inland areas at the same latitude.

Fishing Grounds:

- The **mixing of cold and warm ocean currents** creates some of the world's **richest fishing grounds.** Notable examples include the **Grand Banks around Newfoundland, Canada,** and the **northeastern coast of Japan.** The interaction between warm and cold currents **replenishes oxygen levels and promotes the growth of plankton,** which is the primary food source for fish populations. Consequently, these mixing zones are prime locations for thriving fisheries.

Drizzle and Fog:

- In regions where **cold and warm ocean currents converge,** the weather often becomes **foggy with precipitation** occurring in the form of drizzle. This phenomenon is particularly evident around **Newfoundland,** where the mingling of currents creates persistent fog and light rain.

Climate Influences:

- **Warm and rainy** climates in **tropical and subtropical latitudes,** such as those found in **Florida and Natal.**
- **Cold and dry** climates on the **western margins of subtropical regions,** largely due to the desiccating effects of cold currents.
- **Foggy weather and drizzle** in zones where warm and cold currents mix.
- **Moderate climates** along **western coasts in the subtropics,** influenced by nearby ocean currents.

Tropical Cyclones:

- **Warm ocean currents accumulate heat in tropical waters, providing the energy** necessary for the formation and intensification of tropical cyclones. The warm water acts as a **fuel, driving these powerful weather systems.**

Navigation:

- Ocean currents, often referred to by their **"drift,"** are crucial for navigation. **Surface currents** can reach speeds exceeding **5 knots** (1 knot = approximately 1.8 km/h), while **currents at greater depths tend to be slower,** usually under **0.5 knots.** Ships often follow routes that take advantage of favourable ocean currents and winds. For instance, a ship traveling from Mexico to the Philippines might utilize the North Equatorial Drift, which flows east to west. Conversely, a return journey from the Philippines to Mexico could follow the counter equatorial current in the doldrums, which flows west to east.

Ocean Deposits:

Ocean deposits are the **accumulation of sediments** that settle on the ocean floor, originating from various sources. These sediments are integral to understanding the geological history, environmental conditions, and marine ecosystems. The origins of ocean deposits can be categorized into several sources:

Terrestrial Sediments:

- Terrestrial sediments are materials that **originate from land and are transported to the ocean by rivers, glaciers, wind, and coastal erosion.** These include:
    - **Clastic sediments -** Comprising sand, silt, and clay, these are primarily derived from the weathering and erosion of continental rocks. Rivers carry these sediments to the sea, where they accumulate near coastlines.
    - **Glacial sediments -** Transported by glaciers, these sediments include large boulders, gravels, and finer particles that are deposited when glaciers melt.
    - **Aeolian sediments -** These are fine particles such as dust and sand transported by wind from deserts and other arid regions, eventually settling in the ocean.

Volcanic Sediments:

- Volcanic sediments **originate from volcanic eruptions** and include **ash, pumice, other pyroclastic materials.** These sediments can be distributed over large areas by wind and ocean currents, forming deposits in both shallow and deep-sea environments. Volcanic islands and underwater volcanic activity also contribute to these sediments.

Marine Organisms:

- Marine organisms contribute to ocean deposits through their **skeletal remains and organic matter.** These include:
    - **Biogenic sediments -** Formed from the accumulation of **shells, skeletons,** and other **hard parts of marine organisms** such as foraminifera, diatoms, and radiolarians. These materials can form extensive deposits of calcareous or siliceous ooze.
    - **Organic-rich sediments -** These sediments are **rich in organic matter, derived from the remains of marine plants and animals.** In certain conditions, these sediments can lead to the formation of hydrocarbons over geological time scales.

Extraterrestrial:

- Extraterrestrial sediments are **materials that originate from space, such as micrometeorites and cosmic dust.** Although they **constitute a small fraction of ocean deposits,** they can provide valuable information about the solar system and the Earth's interaction with extraterrestrial materials.

Classification of Ocean Deposits:

Terrigenous Deposits:

- Terrigenous deposits consist of **sediments derived from terrestrial sources,** primarily through rivers, wind, and glaciers. These deposits are typically found near continental margins and include:
  - **Continental shelf deposits -** Sand, gravel, and silt deposited on the continental shelf.
  - **Continental slope and rise deposits -** Fine sediments like mud and clay transported to deeper waters by turbidity currents.

Pelagic Deposits:

- Pelagic deposits are sediments that **accumulate in the deep ocean, far from land.** These deposits are typically **finer and accumulate slowly over time.** They are divided into two main types:
  - **Pelagic mud deposits -** These include red clay, abyssal clay, and siliceous or calcareous ooze.
  - **Pelagic biogenic deposits -** Comprising the remains of marine organisms, these deposits form ooze, which can be calcareous or siliceous depending on the predominant organisms.

Pelagic Mud Deposits:

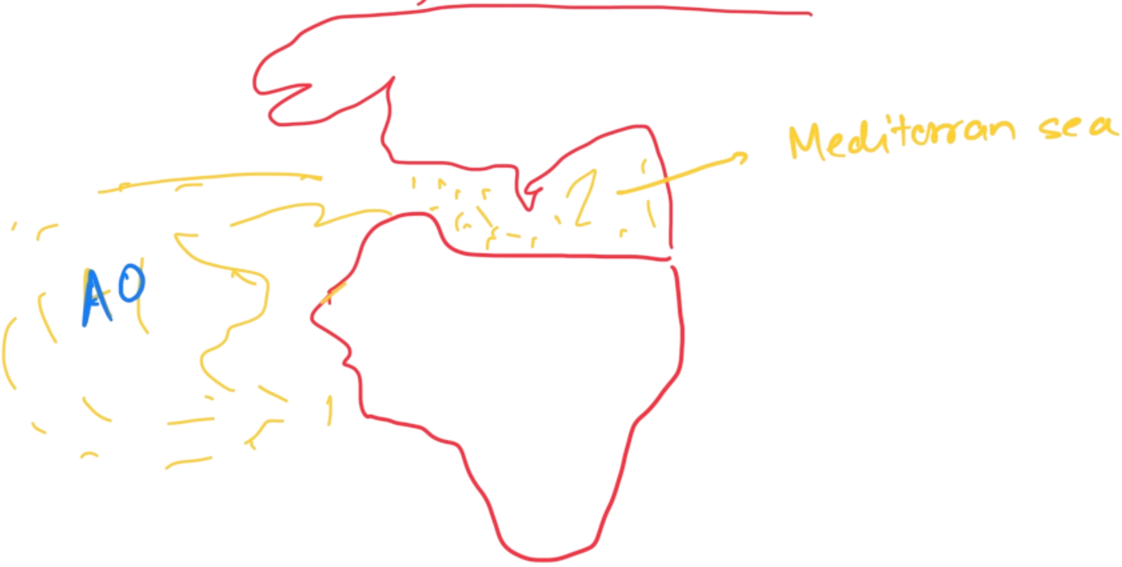
- Pelagic mud deposits are **fine-grained sediments** found in the **deep ocean.** They include:
  - **Red Clay -** Comprising iron and manganese, red clay is abundant in areas far from land where other sediments are minimal.
  - **Abyssal Clay -** Found at great depths, this clay is formed from the slow accumulation of fine particles.
  - **Siliceous Ooze -** Formed from the skeletal remains of siliceous plankton like diatoms and radiolarians, it is found in regions with high biological productivity.
  - **Calcareous Ooze -** Formed from the remains of calcareous organisms like foraminifera and coccolithophores, it is common in shallower parts of the deep ocean where the water is not too acidic.

Ooze:

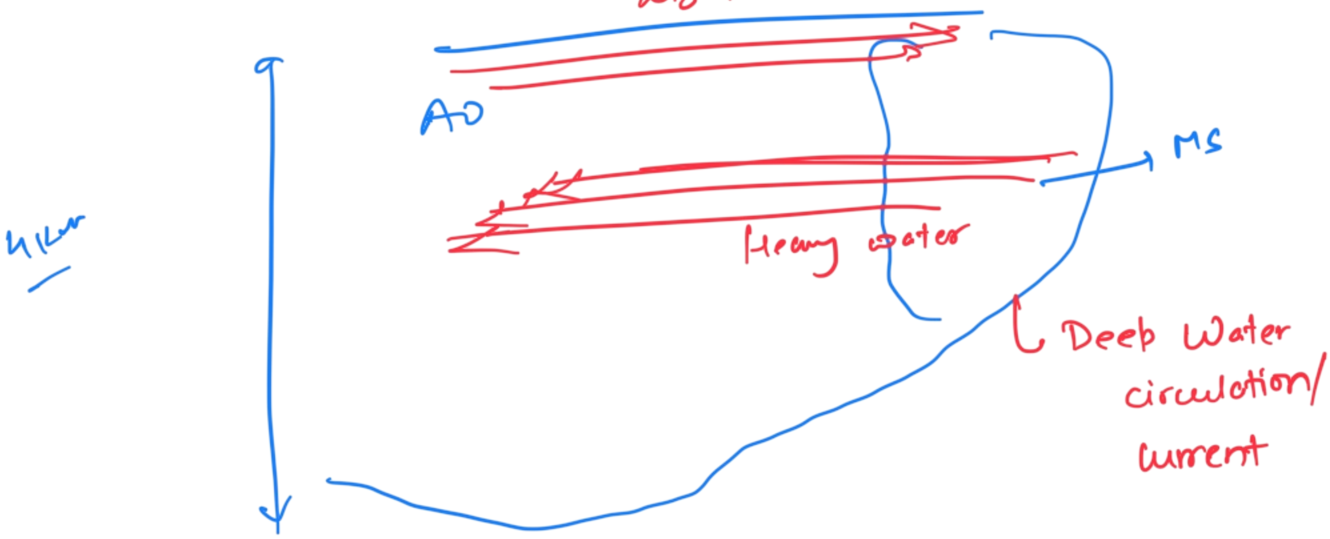
- Ooze is a type of **pelagic deposit rich in the remains of microorganisms.** It covers about **48% of the ocean floor** and is classified into:
  - **Calcareous Ooze -** Contains calcium carbonate and is found in shallower deep-sea regions above the carbonate compensation depth (CCD).
  - **Siliceous Ooze -** Composed of silica and is found in colder, nutrient-rich waters where siliceous organisms dominate.

===== XXXXX =====

Salinity on ocean current



Temp ⇑ Salinity ↑ Density ↑
Heavy water sink



North Atlantic Ocean current

(1) North Equatorial current
(2) Antillca
(3) carriban
(4) Gulf stream
(5) North Atlantic Drift
(6) Norwegian current
(7) Canary current (cold)

South Alta. ocean cu

① South Equa. current
② Goyana
③ Brazilian
④ Antartic circum polar
⑤ Benguella cu.
⑥ Falkland → Polar cold

(8) Labrador current (cold)

(9) Greenland current (cold)

NORTH PACIFIC

North Pacific Equilatorial current

Kuroshio (Japan)

(8) North Pacific current

↳ Alaska

↳ . california cold

Kamchatka current

(6) Okhotsk current /

(7) oyashio Polar cold } current

SOUTH PACIFIC

(1) SPEC

(2) Warm East Aust. current

(3) SPD

(4) Peru / Humbolut - cold curr.

Till This cleared 11 NCERT
↳ Part 1

When India Physical start